

## I. Aircraft

1. The MG-1P Cannot Maintain its Altitude and Drops Suddenly.
  1. Determine whether the obstacle-avoidance radar is correctly installed and working properly:
    - a) If the radar is installed in the right direction, refer to the illustration in the instruction manual for installation.
    - b) Whether the radar rotates normally, and the indicator light on the radar flashes normally.
    - c) Confirm that the radar's altitude stabilization function and the obstacle avoidance function are enabled on the remote controller app.
    - d) After the aircraft takes off, turn the remote controller to the M+ position and gently push the throttle stick to see if the aircraft can descend to the previous height after rising. If yes, then the radar is operating normally.
  2. The radar works normally, but the drone still cannot maintain its altitude and drops suddenly.
    - a) To solve the problem, update the firmware to the latest version.
2. The Radar Enters the Obstacle Avoidance Mode when there are no Obstacles in Front of it.
  1. Cause analysis: The radar falsely triggers obstacle avoidance.
  2. Any firmware problems unresolved will be optimized in the next version.
  3. Solution:
    - a) Inform the user that it is easy to trigger the obstacle avoidance when the terrain is very rugged, and the flight altitude is low. It is recommended to set the aircraft radar stable altitude to  $> 2$  m.
    - b) When conducting edge sweeping in an area with many obstacles, change to the manual operation mode and disable the obstacle avoidance radar.
3. ANR (Application Not Responding) Happens During Re-powering Up When the App is Editing the Route Calling Task.
  1. Cause analysis: It is an app problem which is being optimized.
  2. Solution:
    - a) Inform the users that we are working on it and the app will be optimized. The task should be saved in time if the area is over 100 acres. The issue will be fixed in the next version.
    - b) Ask the user to email a video clip to [apex.zhang@dji.com](mailto:apex.zhang@dji.com).

4. The System Reports the Network is not Available when Logging into the Account, Error Code 00.

Update first and then activate

5. How to Determine if the Compass in the Landing Gear is not Fitted in Reverse

1. Correct installation method: the compass bracket is located on the legs with a vertical ellipse and a horizontal one on each leg. The vertical ellipse is to be facing outward, as shown below:



2. An incorrect compass installation manifests itself as the aircraft being uncontrollable while flying forward/backward or being prone to compass abnormality.

6. Instructions for the Correct Installation of the Obstacle Avoidance Radar on the Landing Gear

1. One end of the radar cable must be close to the spray tank and far away from the lower part of the landing gear. If the radar is incorrectly installed or biased will lead to exceptional height loss.



## 7. How to Identify a GPS Module Installation Error

1. If the drone got wet and its GPS module needs to be replaced, make sure that the GPS module's indicator is installed toward the tail of the drone and the indicator is covered by the white glue.

## 8. ESC Troubleshooting Guide

| Failure type                | Failure phenomenon                                                                                |                                                                                          | Speculative cause                                                                   | Troubleshooting steps and handling methods                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Abnormal ESC communication. | 1. The app prompts abnormal ESC communication.<br>2. The error remains after powering on and off. | The ESC indicator is off.<br>There is no “1356” beep after powering on.                  | ESC board damage                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                             |                                                                                                   | The ESC indicator is off.<br>There is “1356” beep after powering on the motor.           | 485 signal cable is disconnected                                                    | Check if the signal cable is installed properly.<br>Check the 485 cable between the flight controller and the 486 converter board.<br>If the above results are normal, use a signal cable to link the flight controller PPM, 485, and directly link the ESC signal cable socket at the other end to replace the arm signal cable. Power on to see if it is normal; if yes, the arm signal cable is abnormal and needs to be replaced with an arm Y-cable. If problems still exist, please replace the ESC board.                                                                                                            |
|                             |                                                                                                   | ESC indicator flashes yellow.                                                            | The PPM throttle cable is not connected well or is disconnected or short circuited. | Check if the signal cable is installed properly.<br>Check if the PPM throttle cable socket to which the flight controller is linked is installed properly.<br>If the above results are normal, use a signal cable to link the flight controller PPM, 485, and directly link the ESC signal cable socket at the other end to replace the arm signal cable. Power on to see if it is normal; if yes, the arm signal cable is abnormal and needs to be replaced with an arm Y-cable. If problems still exist, please replace the ESC board.                                                                                    |
| Abnormal ESC throttle link. | The app prompts the ESC throttle link anomaly occurs during flight.                               | After powering back on, the fault disappears but then reappears in flight.               | ESC PPM filtering                                                                   | Just replace the ESC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                             |                                                                                                   | The error remains after powering on and off, and the app reports abnormal communication. | Troubleshoot following the same procedure for abnormal ESC communication.           | 1. The ESC indicator flashes red and yellow alternately.<br>Check process: ESC - Y-cable - motor - 485 converter board - main controller - motor LED light.<br>2. The aircraft has crashed into water, and now app reports ESC or throttle cable is abnormal.<br>Check process: ESC - Y-cable - 485 converter board - motor - motor LED light - center board - distribution board - main controller.<br>3. The aircraft fails to update by connecting to the DJI Assistant 2, and the error code indicates abnormal ESC.<br>Check process: Re-update - check ESC - motor - 485 converter board - Y-cable - main controller. |
| ESC self-check error.       | The app prompts abnormal ESC. Faulty ESC indicator flashes red and yellow.                        | The error remains after powering on and off.                                             | The ESC board, or the motor is abnormal.                                            | Check if the ESC board was exposed to moisture or water damage, resulting in patination on the surface. If patina or corrosion is discovered, the ESC must be replaced.<br>If the ESC looks normal, use a multimeter to test the motor resistance. If the resistance is lower than 1KΩ, replace with a new motor and test it.<br>Use a multimeter to test the resistance of the board battery cable to CND, and the motor cable to the VCC. If the tested resistance is less than 10R, the battery board is damaged, and the ESC board needs to be replaced.                                                                |
|                             |                                                                                                   | Failure disappears occasionally after powering on and off.                               | The ESC board or the motor might have                                               | Check if the ESC board is exposed to moisture or water damage, resulting in patination on the surface. If patina or corrosion is discovered, the ESC must be replaced.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

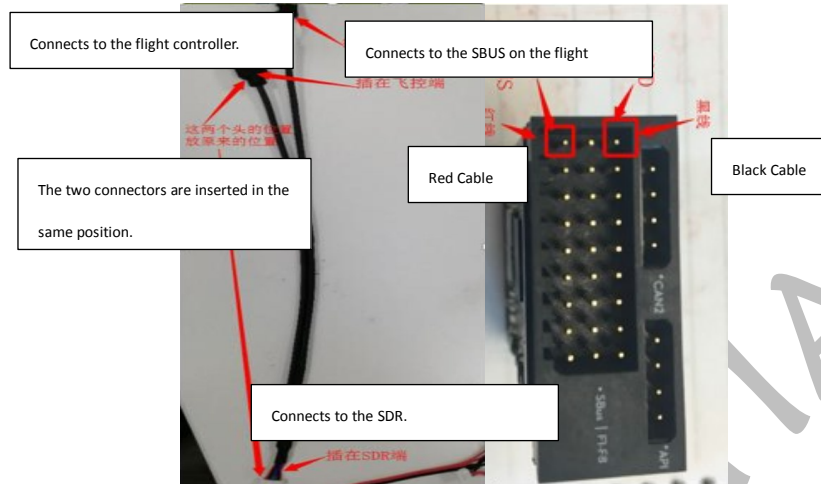
## Troubleshooting Guide for DJI Agricultural Products

|  |  |  |                                                        |                                                                                                                                                                                                                                                                                                  |
|--|--|--|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | been exposed to moisture, or experienced water damage. | <p>If a water stain or moisture is discovered on the surface of the ESC board, power on and test after drying it.</p> <p>If the ESC looks normal, use a multimeter to test the motor resistance. If the resistance is lower than 1K<math>\Omega</math>, replace with a new motor and test it</p> |
|--|--|--|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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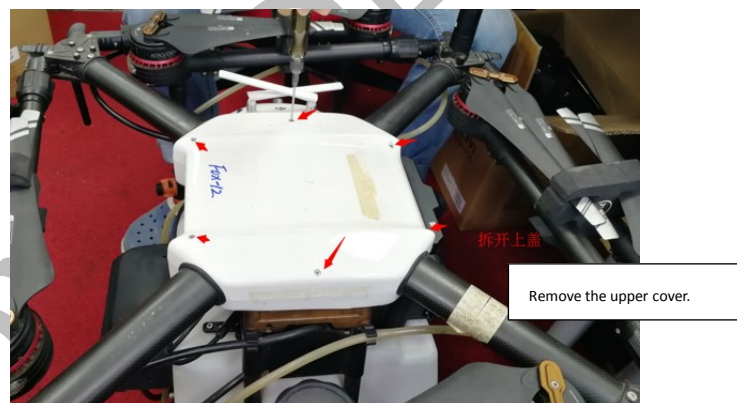
## 9. Operating Instructions for Replacing the Cable Connecting the SDR to the Flight Controller.

1. This cable can solve the MG-1P aircraft out-of-control issue.

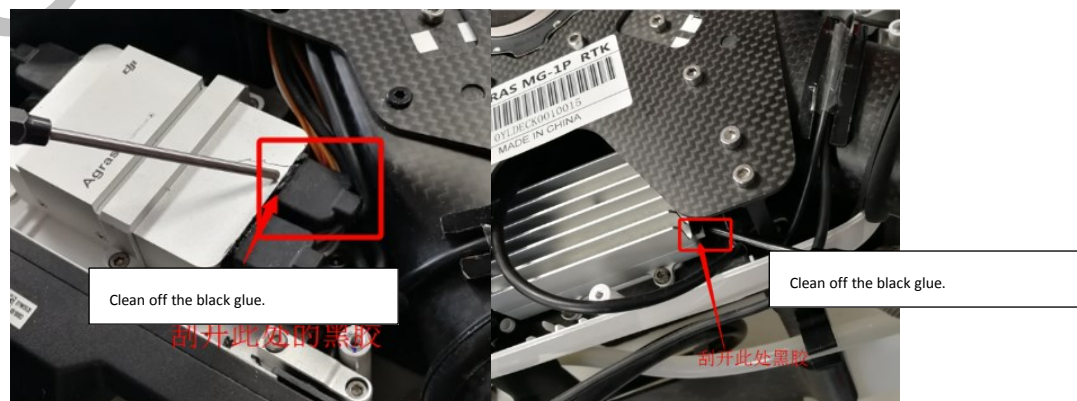


2. SDR cable installation method:

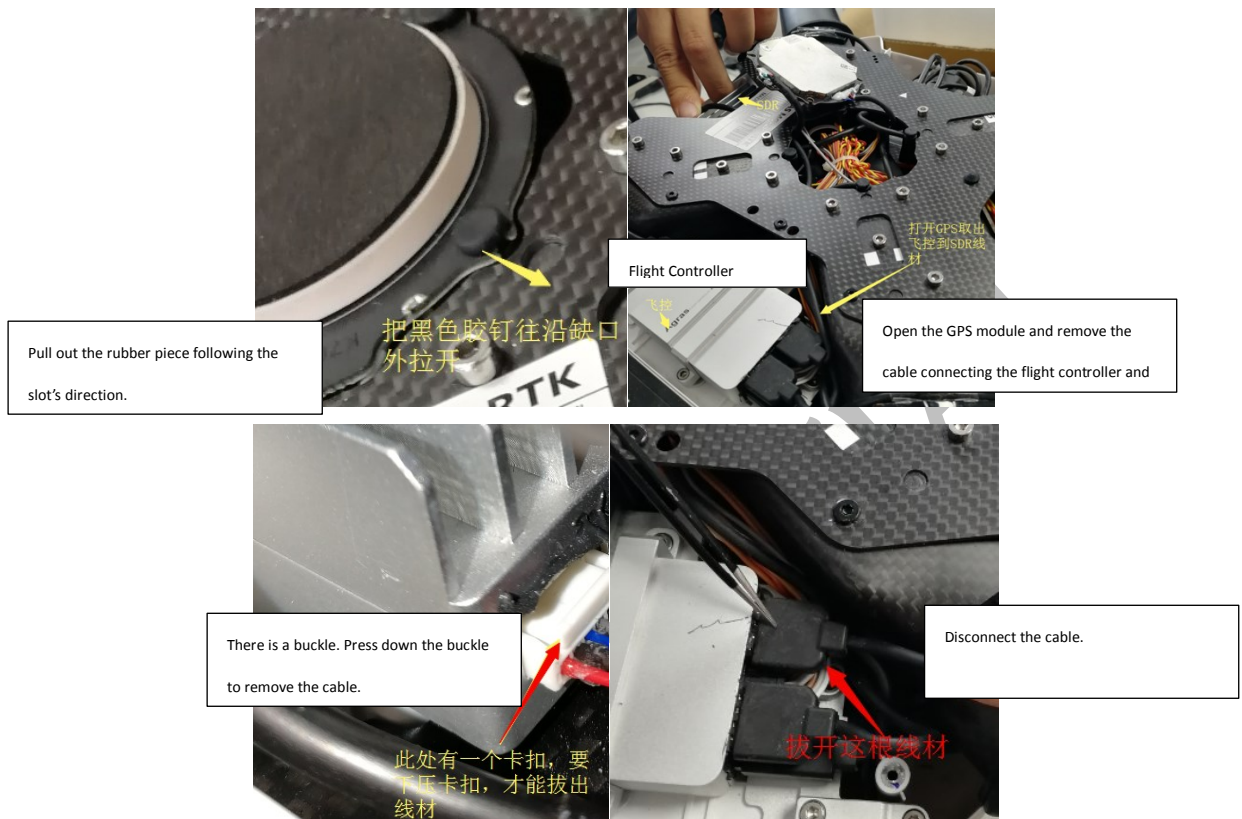
- a) Remove the six screws on the upper cover of the flight controller and put them away.



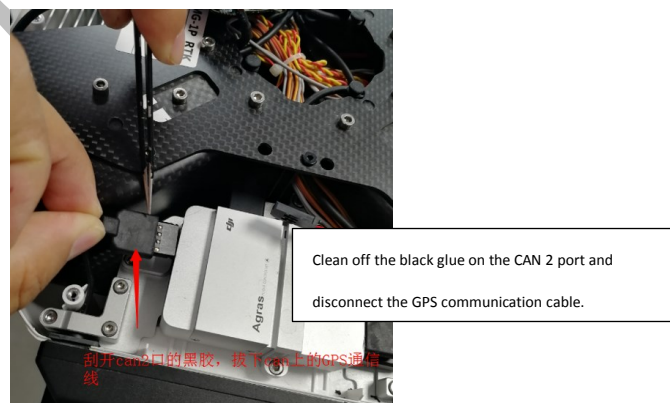
- b) Scratch off the black glue.



- c) Disassemble the rubber sleeve fixed on the side of the GPS module and remove the cable connecting the flight controller to the SDR.

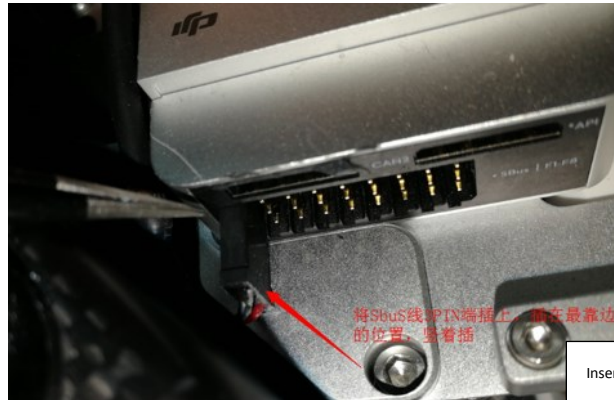


- d) Disconnect the CAN2 port's connector plug.





- e) Replace it with a new cable.



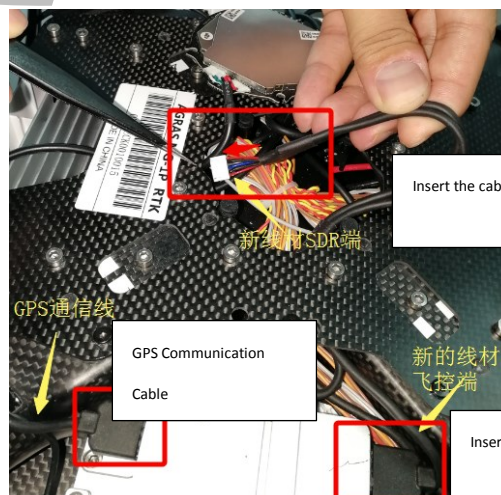
Insert the SBus cable vertically to the far left position as shown.

- f) Connect the flight controller end of the new cable to RF port.



The four parts face upward.

- g) Plug the GPS communication cable back in and pass the new cable through under the carbon fiber board.



Insert the cable to the SDR module.

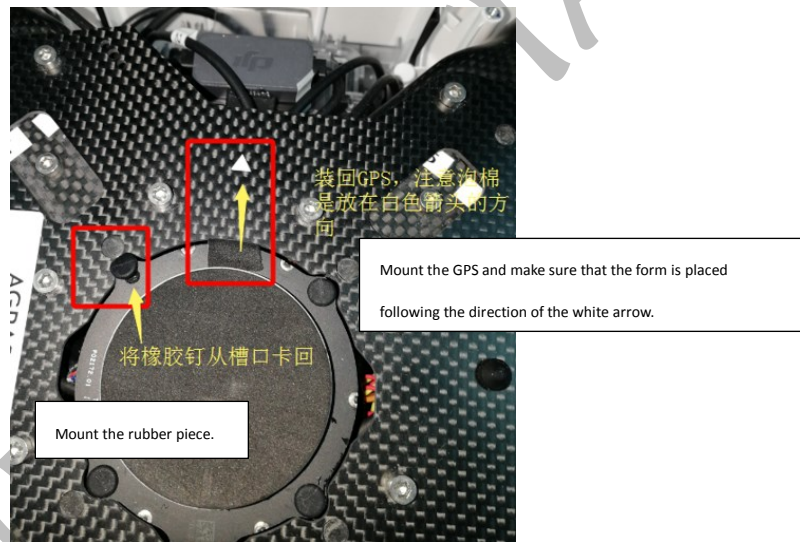
Insert the cable to the flight controller.



- h) Fasten the SDR end of the new cable till you hear a click.

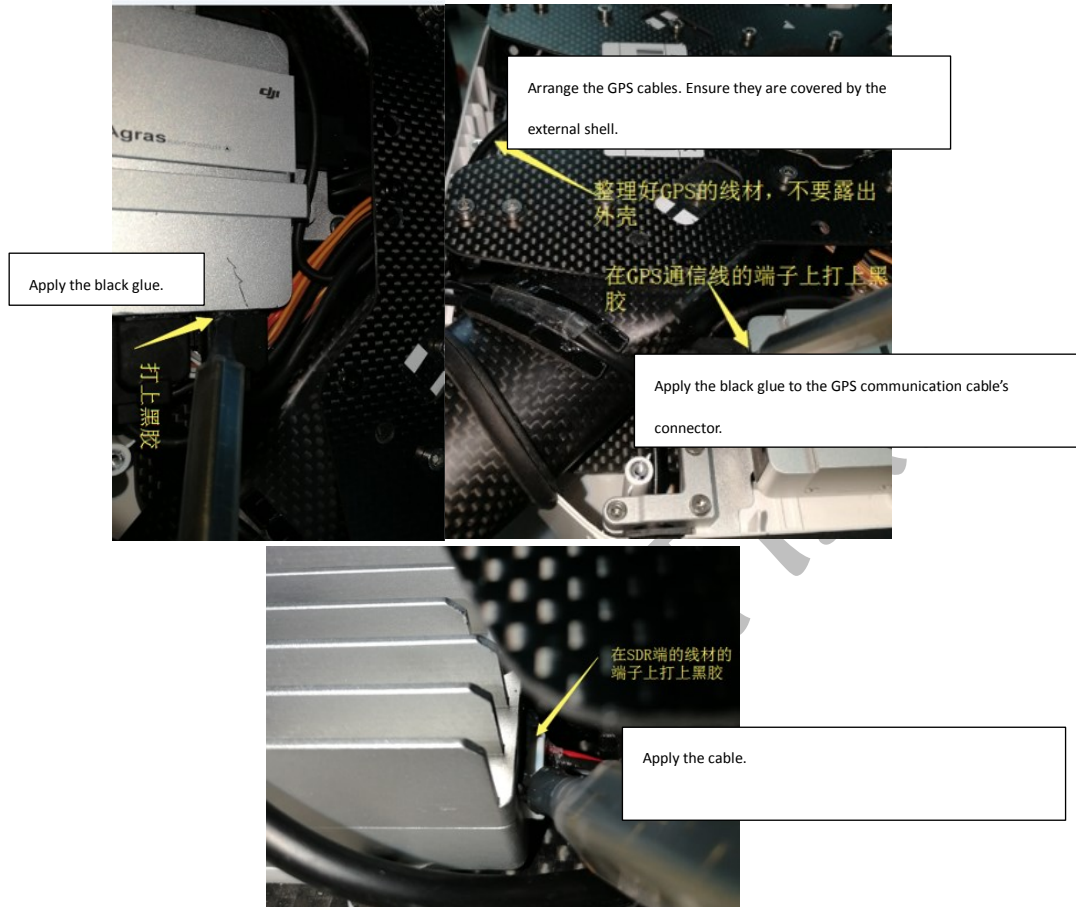


- i) Arrange the cables and reinstall the GPS.



Make sure that the screw is mounted in the original position.  
Arrange the cables.

- j) After installing the cable, apply the black glue to the cable terminal.



- k) Tighten the six screws on the upper cover and reattach the upper shell.



## 10. Troubleshooting Abnormal Intelligent Battery and Flight Controller Communication Error Prompts

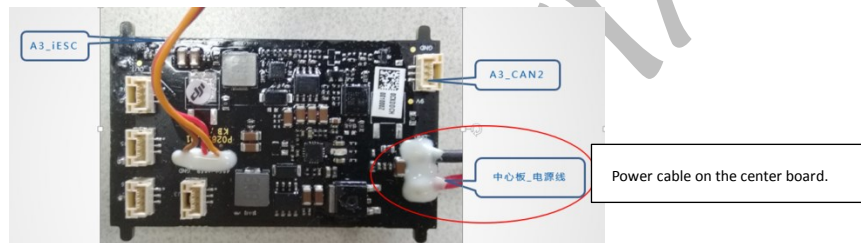
1. When the remote controller prompts that intelligent battery and flight controller have abnormal communication and startup fails:
  - a) Tap the error information on the popup window.
  - b) If there is no error, and the battery level is not displayed in the upper right corner, and only **N/A%** in red or a **red battery percentage** is displayed. Tap the red font.

- c) If only the total voltage is displayed, and the single-cell voltage and the number of discharge cycles are not displayed, then replace one battery first.
  - d) If the error persists when the battery in question is inserted into another aircraft, the problematic battery can be identified:
    - i. If it is a 1P battery, replace a battery communication cable.
    - ii. If it is a 1S battery, it is recommended to return it to the factory for repair (no accessories currently available for repairing 1S batteries).
  2. If the error persists after replacing the battery:
    - a) Replace the aircraft end XT100 connector.
    - b) Pay attention when plugging and unplugging the battery: do not shake it as the XT100 connector will shake when the battery is unplugged. In the long run, the connector not be inserted tightly, resulting in bad communication.
  3. **If the error still cannot be solved by replacing the XT100:**
    - a) For MG-1S model, replace the CAN port on the center board (the CAN port has two sides, with the silver metal side facing down).
    - b) The MG-1A and MG-1P have only one CAN port, and there is no need to replace the CAN port (the CAN port has two sides, with the silver metal side facing down).
    - c) If the error cannot be solved by replacing the CAN port, check the CAN cable (measured with a multimeter). If the CAN cable fails, the battery power will not show, and there will be other errors reported in popup windows, such as magnetic field interference, GPS data exception, compass error, etc.
  4. If the CAN port and the CAN cable function normally, replace the center board:
    - a) If the information in the tapped popup window shows that both the pump and the radar are not connected, the problems are related to the center board. In this case, just replace the center board.
    - b) Even after replacing the center board and solving the communication error, there may still be an error. For example, the pump pressure is abnormal, and the radar is not connected. It can be solved by updating or flashing the firmware.
- [Remarks]:** After replacing an electronic board, such as the ESC or motor, an error may be reported. It is caused by a firmware version mismatch, so they need to be updated. Also, if you only update the aircraft and not the remote controller, then the two versions will be inconsistent, and there may still be an error. So it is necessary to update the firmware for both the aircraft and the remote controller.
5. If error remains after the center board is replaced:
    - a) MG-1S: It is possible to determine whether the error is caused by the barometer by removing the external barometer.
    - b) To remove the barometer, switch the barometer's connection cable with the center board's connection cable, and plug the latter into the barometer socket. Ignore the other barometer cables (For any questions, submit barometer case documentation to the Service Manager).
    - c) This operation is not required for MG-1A and MG-1P without the external barometer.
- [Remarks]:** To remove the barometer, refer to the above instructions for changing the GPS link interface on the MG-1S.
6. If the problem cannot be solved after removing the barometer, it is necessary to replace the power distribution board.

- a) If the cable on the XT100 black board is broken or unsoldered, replace the power distribution board directly (the soldering is difficult and needs to be done with caution).
- b) There was a failure case where the temperature sensor of the battery connector was abnormal. Since this sensor is also located on the XT100 black board and cannot be repaired by itself, the only solution is to replace the power distribution board.

### 11. Troubleshooting and handling of SD card exception issue

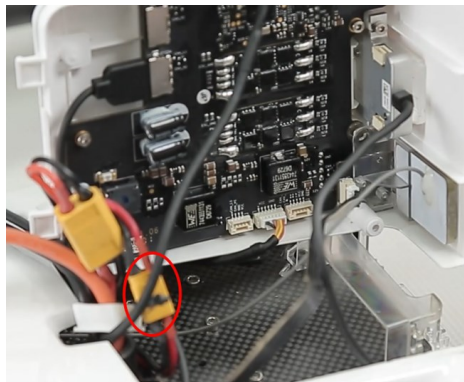
1. Power-off the drone and wait 5 minutes then restart. If an error remains, please do the following:
2. If it happens every time the product is powered on, the SD card is broken. In this case, replace the main controlled.
3. If it is infrequent, the solution is to disconnect the spare battery module.
  - a) Solution: Unplug the cable connecting the resume module to the center board, and wrap the spare battery connector with black tape (for MG-1S only). As showed below:



4. The process of removing the MG-1S resume module
  - a) Reverse the machine and remove the center board's shell, as shown below

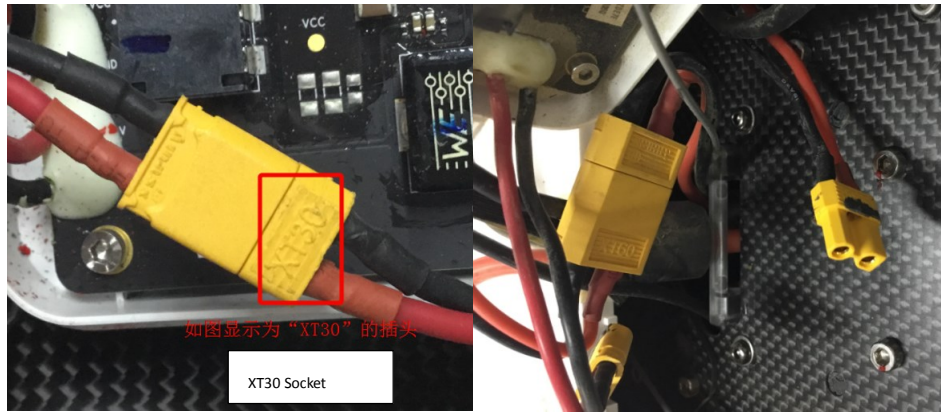


- b) Remove the center board after removing the shell.

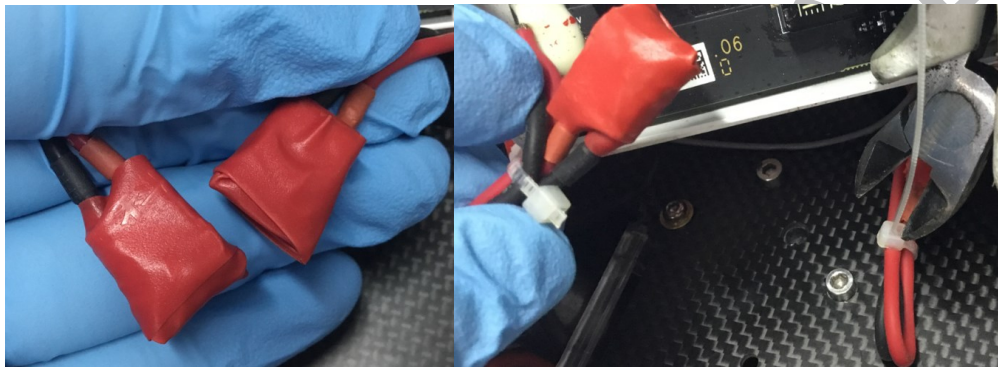


- c) Unplug the power supply cable connecting the center board and resume module.





- d) Use antistatic tape to wrap the ends of the connector, fix the cable with a cable tie, and cut off the extra length.

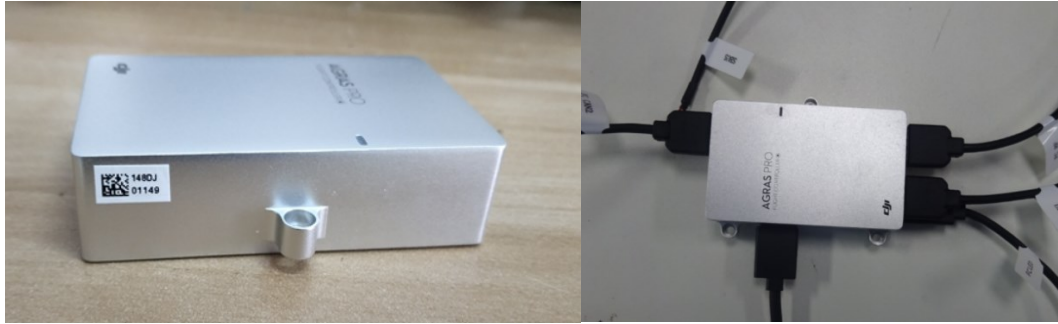


- e) Reattach the rear shell.



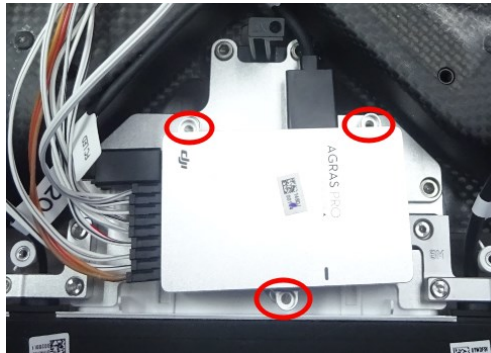
## 12. Installation Instructions for the Optimized MG-1P/1SA/1PRTK New Controller

1. The serial number of the main controller after delivery starts with 148. This flight controller is an optimized version of the old one. However, if this flight controller is used, it is necessary to add the main control lower fastener to attach the main controller.
2. If the new controller is involved in replacement, the old flight controller and the old fastener must be sent back. If the old fastener of the main controller is in good appearance, the new main controller lower fastener will be replaced for free; if the old fastener is damaged, the customer is required to replace it with the new version of main controller lower fastener. The figures below shows where to install the cables on the new controller (left) and the main controller (right):

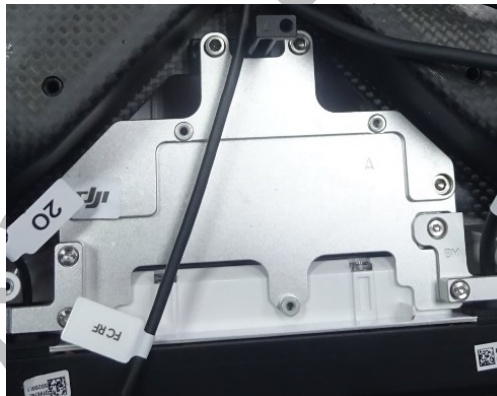


### 3. Accessories installed in the new main controller and installation operations:

- a) The figure shows the position of the three screws securing the main controller.



- b) New main controller's mounting piece.



### 13. IMU Error 10 Troubleshooting Method

1. Solution: Restart, place the aircraft on the level ground for a while. If the initial attitude test cannot be passed within 30s, first check the GPS cable or replace the GPS module. Otherwise, just replace the flight controller.
2. Troubleshooting process: Restart - multi-directional calibration – check the GPS or replace - replace the main controller.

### 14. Troubleshooting and Handling Measures for Motor Overheating

1. Check if the ESC was exposed to water damage or moisture. The ESC has a liquid contact

indicator in it. If the indicator has turned red, then the ESC probably has water damage. Solution: Replace the ESC.

2. Check if the motor has crash damage and if the motor angle is correctly tilted 3 degrees. Solution: Adjust the angle and replace the motor.
3. Check the motor base for short circuits or foreign objects, and check if the screws are loose.



4. Check if the screws on the carbon tube and Y-tee part touch the Y-cable and if the Y-cable is damaged or short-circuited. Solution: Replace the Y-cable.
5. Check if the circuit is secure. Solution: Test by re-plugging or changing the cables alternately.
6. If the aircraft is exposed to water damage and severe moisture, it is necessary to replace all the electronic boards and all the cables, including power cables and Y-cables. (Recently there are many cases where Y-cables are not replaced after the aircraft was exposed to water damage, which led to the motors overheating and crashes).

[Remarks]: If the components need to be replaced, the screw glue needs to be applied after the aircraft is repaired.

Picture of screw glue:



### 15. Troubleshooting Method for Aircraft Spinning Out of Control while Taking Off

**Understand the geographical environment where it took place:**

Could the aircraft make normal takeoffs before? Did it happen during operation or under manual control? What about the firmware version information? Was there a crash?

**Troubleshooting process for general scenarios:**

1. Determine if the landing gear and spray tank are properly installed.



2. Check the motor inclination angle and propeller installation/condition. Third-party propellers or worn propellers may also cause the aircraft to vibrate and thus spin out of control. Link to motor angle adjustment tutorial: <https://share.weiyun.com/5ydYyhJ> Password: ndvses
3. Multi-directional IMU calibration. Tutorial video: <https://v.qq.com/x/page/a0718o8zpav.html>
4. Calibrate the compass.
5. Ensure that the aircraft is flying without magnetic interference and same-frequency interference.
6. Check the rotational direction. If the aircraft rotates clockwise, check the No. 2468 motor's inclination angle and the ESD welding rod. If it rotates counterclockwise, check the No. 1357 motor's inclination angle and the ESD welding rod.
7. If none of the above methods can solve the problem, please export the flight controller data and send to the service manager.

[Remarks]: Link to the flight controller data export guide: <https://share.weiyun.com/5EPE1To>  
Password: ecfvkt

**Special scenarios: If out of control spinning occurs after the customer tests their self-repair.**

**Troubleshoot according to the above process. If it is ineffective (if disassembly is involved, guide the customer to contact a qualified repair agent for troubleshooting):**

1. Check if the compass is installed in the correct direction. Compass installation instructions link: <https://share.weiyun.com/5hQyD4p> Password: nhe5p8
2. Verify if the main controller is installed securely or in place, check for vibration caused by incomplete 3M glue after the main controller is replaced.
3. Remove the upper cover of the aircraft and check the GPS installation.

4. Check if the soldering between the ESC and the motor is firm and if there is any cold solder joint or non-standard soldering.

## 16. Aircraft Flying in One Direction Only While Taking Off

1. Check if the GPS and compass wiring is correct.
2. Check if the compass is reversed.
3. Check if the radar is reversed.
4. Check the satellite search status and replace the GPS module if necessary.
5. Check the motor inclination angle.
6. Recalibrate the remote controller stick.
7. Re-update the aircraft kit.
8. There may be an error with the main controller's internal speed value, replace the main controller.

MG drone series video tutorial:

[http://i.youku.com/i/UMTKxOTMxODQwNA==?spm=a2h0j.11185381.module\\_basic\\_sub.A](http://i.youku.com/i/UMTKxOTMxODQwNA==?spm=a2h0j.11185381.module_basic_sub.A)

## II. Remote Controller or App

### 17. The Remote Controller Cannot be Used After the Network Card is Inserted

1. Check if the network card can be connected to the Internet.
2. Connect the remote controller to a third-party wireless network (such as Wi-Fi or mobile hotspot), and make sure the wireless network function of the remote controller is in good condition. Alternately test by installing the network card on other remote controllers to see if it is effective.

## 18. MG-1P Remote Controller cannot Connect to DJI Assistant 2 for MG

1. Use a different computer.
2. Try again after restarting the DJI Assistant 2. If DJI Assistant 2 is not recognized, uninstall and reinstall the software.
3. Re-enter into the main interface of the remote controller and re-test. If it fails, open the remote controller manager to see if there is an unknown certification mark. If it fails again, reset the remote controller to default factory settings.
4. Press WIN+R on the computer, type devmgmt.msc in the pop-up box, and press Enter to view the USB Composite Device.
5. Update with the latest firmware package.

**[Remarks]:** For MG-1P, the remote controller can only be updated after connecting to DJI Assistant 2 for MG. Listen for a beep when the remote controller is connected to the computer which shows that the connection is successful. (DJI Assistant 2 for MG download link: <https://www.dji.com/cn/mg-1p/info#downloads>) Remote controller LOG data export link: C:\Users\用户\AppData\Roaming\DJI Assistant 2\MG\log

## 19. MG-1P Remote Controller App Prompts Battery and Main Controller

### Communication is Abnormal

1. Test by installing the battery on another aircraft to confirm if it is a battery problem.
  2. If it is confirmed to be a battery problem:
    - a) End customer: Please contact the dealer to replace the battery communication cable.
    - b) Dealer: Please refer to the method provided in the video below for replacement of the battery communication cable:  
[http://v.youku.com/v\\_show/id\\_XMzM4NjMwMzg0NA==.html?spm=a2hzp.8253869.0.0](http://v.youku.com/v_show/id_XMzM4NjMwMzg0NA==.html?spm=a2hzp.8253869.0.0)
- If none of the above methods can solve the problem, please contact the regional service manager to seek a solution.

## 20. 800 Error Reported during MG-1P Remote Controller Firmware Update

1. Try to reset the factory default settings and format the SD card before updating again.
2. Stop the update processes in the system-application and try to update again.
3. If the problem still exists, please export the tombstone file & rc log and feedback to the regional service manager.

**[Remarks]:** If the remote controller is reset to default factory settings, the flight statistics that have not been uploaded will be deleted.

21. Remote Controller Panel Displaying Disconnection, Image Transmission Works, but the Aircraft cannot be Controlled

The aircraft's and remote controller's indicators are green, showing that they are linked normally, but remote controller's LCD shows that is disconnected, image transmission works, but the aircraft cannot be controlled.

1. Update the aircraft to the latest firmware version.
2. Replace the cable connecting the SDR module and flight controller.

22. The Remote Controller cannot Access the Internet and cannot Connect to the Management Platform

1. ZTE network card
  - a) Debug the network card according to the knowledge base.
  - b) If the network card has flashed blue/green but still cannot access the internet. It is known that the 5003 version has a problem with DNS settings and it can be solved by updating the firmware version.
2. Huawei network card
  - a) Huawei network card does not need to be debugged and has the same frequency band as a ZTE network card. In the normal service condition, only the leftmost indicator light is on.
  - b) If the remote controller cannot access the Internet after the network card is inserted, please check the remote controller's firmware version. The minimum remote controller firmware version supported by Huawei network card is 0048.

23. The New Remote Controller is in Good Condition, and the Control Stick has been Calibrated Several Times, but the RC Keeps Beeping

1. Check the operating temperature. If the temperature is above 70°C before taking off, the aircraft will sound an alarm.
2. Check the remaining power when the power is less than 10% an alarm will trigger.
3. If the stick is not in the center position when the remote controller is started, enter the app and conduct stick calibration.

24. Satellite Search Error in Planning Mode on MG-1P Remote Controller

MG-1P remote controller may have a satellite search error in Planning Mode due to a poorly connected GPS antenna.

Root cause: The MG-1P remote controller may not be able to search for satellites when planning a plot due to a GPS antenna problem.

Solution: Contact the nearest dealers to replace the display module or send the product back to DJI.

## 25. App Prompts “Cannot take off due to magnetic interference”

Root cause: The inductor in GPS module is burnt because GPS cable and the compass were exposed to water or moisture.

Solution: Both the compass and GPS module need to be replaced.

## III. Battery

### 26. Reducing the Risk that the Mg-1P Battery May Over-discharge

1. If the battery firmware is V1.1.3, the battery can be mounted to the aircraft to update to the latest version (V1.3.2.6).
    - a) The battery is connected to the aircraft, and the remote controller is already linked. The “BAT” can be displayed in the remote controller’s LCD screen by continuously tapping “Remote controller app firmware version - Aircraft version” 10 times.
  2. If the battery version is up to date (V1.3.2.6), it is possible to normally use batteries without faults and to update to the latest version of the app (this version has the additional function of updating the battery firmware to the latest version by connecting the battery to the aircraft).
27. If a Battery has been Over-discharged and cannot be Turned on or Charged, Return it to the Factory for Repair. The App Reports a Battery Connector Problem

The app reports a battery connector problem: the sensor is abnormal, or battery connector’s temperature is too high.

1. Root cause: the battery connector’s temperature is too high, the battery connector’s temperature sensor is abnormal, or there is a connection problem between the center board and the aircraft’s battery connector.
2. Version exception: all versions.
3. Solution:
  - a) If the app reports the battery connector’s temperature sensor is abnormal, the battery connector’s sensor may be broken or the connection between the center board and the battery connector may be loose or broken; the troubleshooting method is as follows:
    - i. Replace the connection between the center board and the battery connector. If the warning “Battery connector temperature sensor abnormal” disappears, the connector is broken. Otherwise, continue to check.
    - ii. Replace the battery connector. If the report “Battery connector temperature sensor abnormal” disappears, the sensor is broken. Otherwise, contact the regional service manager for assistance.



- b) If the app prompts that the battery connector temperature is too high, the battery connector sensor may be broken, or the battery connector turns black due to long use time usage.
  - i. If the app reports “Battery connector too hot” as soon as the aircraft is powered on, the battery connector may have a broken sensor. Please replace the battery connector.
  - ii. If the app reports “Battery connector too hot” after a period, the battery connector turns black due to long term usage. So replace the battery connector.

**[Remarks]:** When “Battery connector temperature sensor abnormal” and “Battery connector too hot” are reported at the same time, please first check “Battery connector temperature sensor abnormal”.

## 28. Progress Bar Stuck at 0 when Updating the Batteries Firmware on the MG-1P Aircraft

- 1. Cause analysis: 1860 did not detect that the battery version was inconsistent, but the app did. So the app prompts to update, but the 1860 did not respond to this update action.
- 2. Firmware problem: it will be optimized in the next version.
- 3. Solution:
  - a) Restart the aircraft and the remote controller.
  - b) If the progress bar is still stuck at 0% when update is tapped after several restarts, it is recommended to update using DJI Assistant 2.

## 29. The Battery Shows that the Update was Successful, but it Prompts to Update after being Restarted

- 1. Cause analysis: it is possible that the prompt is triggered because the aircraft is not completely powered off after the user updates the battery, or the battery was swapped for one that wasn't updated.
- 2. Solution:
  - a) Verify if the update was prompted when the user put in an un-updated battery. Inform the user that the batteries are independent and each battery needs to be updated.
  - b) Tap the aircraft firmware on the remote controller 10 times to shown the battery firmware package information on the LCD screen to determine whether the 1100 version is 01.03.04.02.
  - c) When the battery firmware update fails, the battery may be in loader state (displayed by two or three of the battery's LEDs always being on). Re-use DJI Assistant 2 to update the battery.

### 30. Abnormal Battery Communication

1. If the battery problem is confirmed, the communication cable can be replaced for the 1P battery, but the 1S battery can only be returned to the factory for repair. For details, please contact the DJI technical staff at 400-700-0303.
2. If the battery is normally functioning, clean the yellow power connector at the aircraft end.
3. Check if the CAN cable connecting the center board to the flight controller is tight and in good condition.
4. If the replacing the power connector cannot solve the problem, replace the rear shell component (center board).

### 31. The App Prompts that the Temperature is too High

1. Clean the yellow power connector on the aircraft, and check whether the power connector is oxidized or corroded.
2. Update the aircraft's firmware to the latest version as older version may cause false reports.
3. Replace the yellow power connector on the aircraft with a special XT100-L connector.
4. If the above methods cannot solve the problem, replace the smart connector module.

### 32. MG-1P Battery Balance Connector

1. For the sake of safety (direct output via the cell balance cable), the balance cable fuse is added to the MG-1P. When the current is too large, the fuse will be blown. So it is not allowed to draw power from the balance cable to supply other devices.
2. For the cell balance, removing several batteries will cause the battery voltage to be unbalanced, and even crash if it is severe.

### 33. Dealer determines guidelines for the MG-1P battery communication cable

1. If the appearance of the battery communication cable is damaged, it will be determined as the customer's responsibility.
2. If the battery communication cable is in good condition but is dysfunctional, it can be processed within warranty. (The warranty period of a battery communication cable is the same as battery's warranty period).
3. If the battery communication cable looks good but the cable is malfunctioning, and it is out-of-warranty, the customer is liable for the issue. (The warranty period of a battery communication cable is consistent with the battery's warranty period).

[Remarks]: Inspection will be arranged for the damaged parts returned. If the above conditions are not met, no compensation will be given for the corresponding materials and labor costs.

34. Abnormal Batteries Mainly have the Following Problems:

1. The communication cable is broken.
2. Deformed battery/shell.
3. Blown connection board fuse.
4. Abnormal battery communication.

35. Common battery damage.

1. Bottom shell damage



2. Upper shell damage



3. Dented cell body.



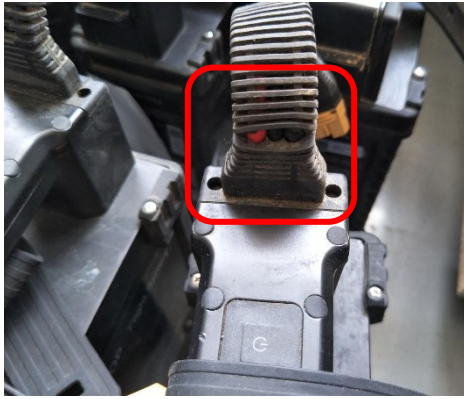
4. Charge socket damage.



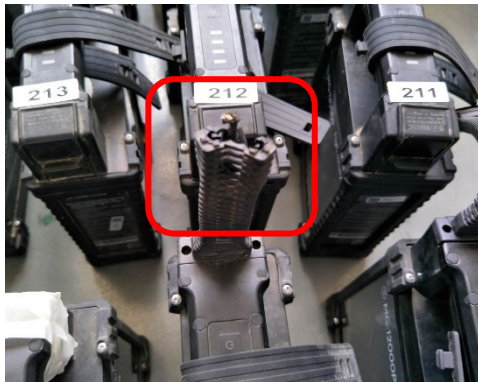
5. Communication cable damage.



6. Communication cable damage.



7. Broken communication cable.



8. Cause of Damage: Non-standard storage and transportation by the user can cause damage to the battery itself or related components.

9. Suggestion:

- a) When the battery is centrally stored, it is recommended that 4 to 6 batteries form one unit, and each storage unit is separated by more than 30 cm.
- b) Keep the battery storage location away from environments with large items being frequently moved to prevent accidental damage.
- c) It is advisable to use temperature-resistant or flame-retardant boxes with a cover to store the batteries. Inside the box, the shock-proof and anti-tilt sponge in the original packing containers for the batteries should be used.
- d) Do not pull the communication cable to lift the battery.
- e) Do not over-bend communication cables.

Avoid overlapping the battery charge communication cables on the box edge to prevent accidental crushing and short-circuiting.

### 36. Battery storage

1. Keep away from heat sources (inside the cars exposed to direct sunlight, fire, hot furnaces, etc.). The ideal storage temperature for batteries is 22°C - 28°C.
2. Keep dry, and do not place the batteries in water or where they may be exposed to water.
3. Keep the batteries away from children.
4. Do not mechanically hit, crush, drop, artificially short-circuit, or pierce the batteries.

5. Keep away from metal during transporting, and do not store or transport the batteries together with metal necklaces, hair clips or other metal objects.
6. If a battery is not used for more than 7 days, discharge it to 30%-50% for storage, which will greatly prolong the battery life.

If a battery is not used for a long time, it is recommended to have the battery over 65% charged.

Turn on the Self Discharge in Storage mode, and store it in the dedicated battery box. **Recharge and discharge once about every 3 months to keep batteries active.** Do not store the batteries for a long time after discharge to avoid over discharging the batteries leaving them unusable.

### 37. Burning Caused by the Improper Installation of the Battery Communication

#### Cable Connector

Problem description: Burning may be caused by the insulating tape melting in high temperature due to the failure to fasten the screws or to residual insulating tape between the copper posts during replacement of the MG-12000P's communication cables.

Solution:

1. Be sure to clean off the insulating glue when removing the old cable.
2. Make sure the screws are fastened during installation.

Image:



## IV. Charger

### 38. MG Hex Charger is only Charging One Battery

1. The logic of the MG Hex Charger: It can charge six batteries, but it will charge two batteries to the same power level first, then the other two batteries.
2. The charger charges two batteries at the same time. If only one battery is charging, check the connection between the batteries and the charger.
3. If it is determined to be a problem with the communication cable, charge the two batteries to the same power level first, then the other two batteries.
4. If none of the above methods can solve the problem, send it for repair.

### 39. Troubleshooting the MG Six-channel Smart Charger's Alarm

1. Cause analysis:
  - a) Connection problem.
  - b) High battery temperature.
  - c) Battery over-discharging.



- d) Charger malfunction.
2. Solution:
  - a) Check the battery cable, re-plug or replace with another socket.
  - b) Observe the battery temperature. If it is too high, put it in a cool, well-ventilated place for a while before charging.
  - c) Connect the battery to the aircraft, and check the state of the battery cell in the app for any cell abnormality.
  - d) If charging fails after more than one battery is changed, ask the user to take a video of the whole operation process and provide the charger's serial number to the regional service manager.

#### 40. MG Hex Smart Balance Charger

Failure phenomenon: MG Hex Charger sounds a false alarm about the fan (random failure)

Solution: Update the charger's firmware (see attachment for details)

Update tool download link: <https://share.weiyun.com/5LSpSd0> Password: 5av3mk

#### 41. MG Smart Charger Online Update Guide

1. Problem: When the MG smart balance charger is charging an old battery, the voltage may jump to above 4.2V causing the charging to stop. So the customer may think that the charger cannot fully charge a battery.
2. Solution: Update the charger's firmware. Link to firmware and update method:  
<https://share.weiyun.com/5OZjbr8> Password: hu2fix

Image:



## V. Spraying System

42. The Spraying System can Operate Normally, but the App Frequently Warns that there is Air in the Pipeline, Traffic Data is 0 or the Data Value Jumps too much in the App

1. Root cause: With the increased service life and gradual wear and tear on the pump, the spraying system is not sensitive to pipeline air detection triggering a false alarm.
2. Version exception: all historical versions.
3. Solution:
  - a) Strictly follow the flow calibration instructions on page 45 of the User Manual, and perform a flow calibration or perform the calibration to determine whether there is air in the pipeline. (Manual download link: <https://www.dji.com/cn/mg-1p/info#downloads>)

[Remarks]: The flow calibration actually involves calibration of the spraying flow and calibration for determining whether there is air in the pipeline. The latter is just used to determine whether air enters the pipe.

43. The App Warns that there is Air in the Pipeline, the Spray Pump Rotates Normally but Fails to Spray Water

1. Root cause: There is air in the pipeline which prevents the pump from spraying water.
2. Version exception: all historical versions
3. Solution:
  - a) Vent the air. It is best to remove the four pressure relief valves for complete air venting. When the nozzles are all stable, the pressure relief valve is tightened.

44. During RTH, the App Warns that there is Air in the Pipeline. After Returning to the Home Point, it Finds that the Spray Tank is Empty

1. Root cause: The liquid level meter is damaged or the spraying volume is not accurate.
2. Problem version: All historical versions
3. Solution:
  - a) Put one hand into the spray tank and toggle the level gauge back and forth every 2 seconds. Open the app and go to the “Spraying system setting page” / “Spraying system real-time data”. If the level gauge data displays “with liquid” and “without liquid” alternately, the level gauge is normal; also a voice prompt should be heard stating “liquid fully sprayed”. If the level gauge data does not change, the level gauge is

damaged and should be replaced. If the level gauge is normal, proceed to the following steps.

- b) Use a standard measuring cup to perform a flow calibration in strict accordance with the manual.

#### 45. When Removing the Pressure Relief Valve for Air Venting, the App Always Warns that there is Air in the Pipeline

1. Root cause: The working condition when the pressure relief valve is removed for air venting is the same as the condition in which air enters the pipe. It is impossible to distinguish whether air enters the pipe or the pressure relief valve is open.
2. Version exception: all historical versions
3. Solution:
  - a) This is a normal phenomenon. Remove the pressure relief valves for air venting. When the nozzles are all stable, the pressure relief valve is tightened. Then the app prompt notifying of air in the pipeline will disappear.

#### 46. There is Deviation with the Spraying Volume

1. Root cause: The aging of pumps, pipes, nozzles, etc. will gradually affect the amount of liquid sprayed.
2. Version exception: all historical versions.
3. Solution: Perform a flow calibration again.

#### 47. Quick Check of Center Board Link

1. Manually turn on spraying and flip the impeller to adjust the flow rate. If the spraying system responds to instruction normally, the link from the remote controller to the spraying system is normal. If otherwise, it is abnormal.
2. After spraying is turned on, open the app and go to “Spraying system setting page” / “Spraying system real-time data”. If the real-time data of the spraying system can be seen, the link from the center board to the app is normal.
3. Tap the battery icon on the app to open the “Smart Battery Settings” page. If the real-time information of the smart battery, the link from the app to the battery is normal.

#### 48. Sensor Normal Range Table

| Sensor                                      | Static  | Spraying | Remarks                                                                                                                                                                                                                                                      |
|---------------------------------------------|---------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pressure sensor value (kg/cm <sup>2</sup> ) | 0.4~1.8 | 1.8~5.5  | The pressure sensor's value may be temporarily high during flow calibration, which is normal: during normal spraying, if the value is above 6 kg for more than 5 seconds, it is determined to be a failure, which may be caused by the being nozzle blocked. |

## Troubleshooting Guide for DJI Agricultural Products

|                        |   |           |                                                                                                                                                                                                                                                                            |
|------------------------|---|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Speed value<br>(r/min) | 0 | 500~1,700 | The spinning speed may be temporarily high during flow calibration, which is normal: during normal spraying, if the speed is above 2,200r/min, it may be because the pump is aged. If the pressure sensor's value is normal, it can be used as usual without any problems. |
|------------------------|---|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

[Remarks]: If there is a difference between the values of the left and right pumps, a flow calibration may be performed. If the difference still exists, it may be caused by the difference between the pumps. As long as the values are within a reasonable range, it is normal.

## VI. Agricultural Service Platform

### 49. Incorrect Flight Statistics

1. Incorrect flight statistics include: The acreage is 0, incomplete operation records, and missing operation records for a certain period of time. The following are the basic steps and solutions to solve the problem:
  - a) V1.3.26: First guide customers to use the remote controller to connect to the network. The agricultural service data will be automatically uploaded when connected to Wi-Fi.
  - b) V1.4.5: Guide customers to use the remote controller to connect to the Wi-Fi and upload operation records in time. For users who have reset to default factory settings, inform them that the resetting will clear all the previous operation records.
  - c) V1.5.7: Guide customers to update the remote controller app to the latest version and try to upload data.
  - d) V1.6.26: Guide customers to connect the remote controller to the network while ensuring that the Wi-Fi network is in good condition. Wait and complete the data upload.

[Remarks]: Make sure that the user's data synchronization switch on the agricultural management platform is on. If it is off during operation, the flight records before the switch is on will not be uploaded even if the user turns on the switch later. If none of the above methods can effectively solve the customer's problem, please contact the service manager to seek a solution.

## VII. Others

### 50. Prompt of No-fly zone for MG Drone

1. Know the specific location where flying is restricted and check if the aircraft is in a no-fly zone.
  - a) If it is in the no-fly zone, inform the customer of the process to unlock the zone, and contact the regional service manager.

- b) If it is not in a no-fly zone, judge whether the aircraft is within 5 km of a no-fly zone border.
- c) If the aircraft is outside the 5 km range, i.e., no-fly zone, and it prompts flying restriction, try to reset the remote controller to default factory settings.

[Remarks]: Flying will be banned within 5 km of the border, and the ban cannot be lifted.

### 51. MG Data Analysis Tutorial

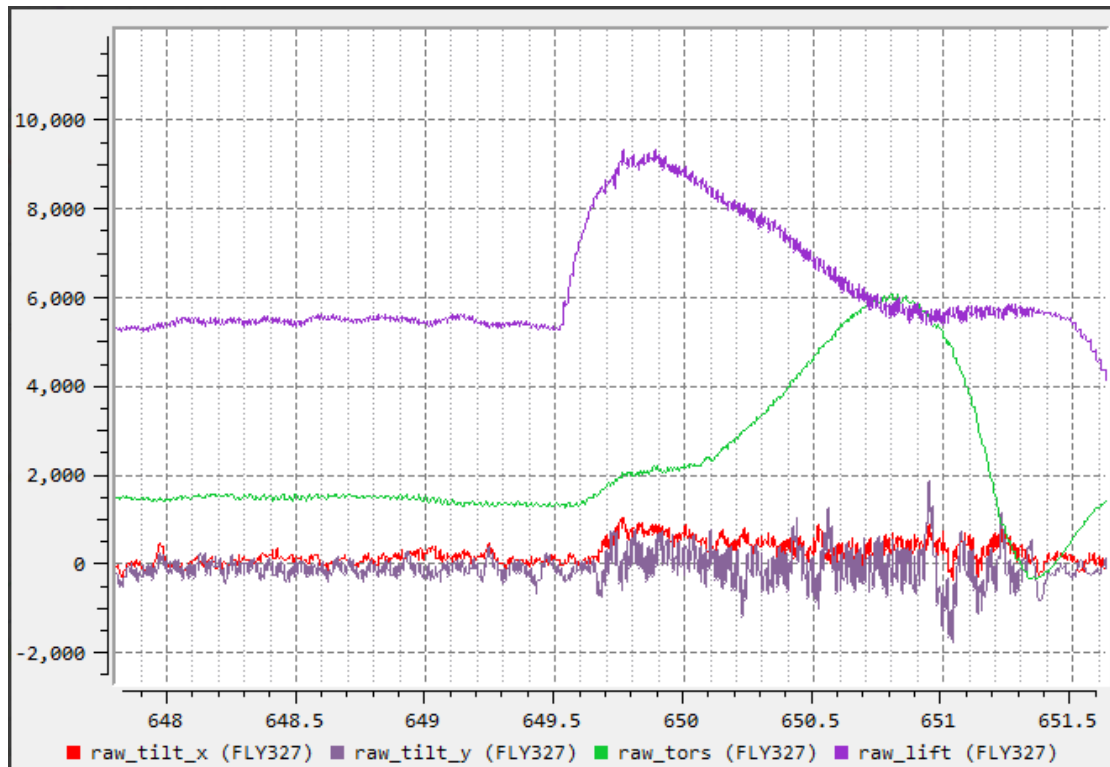
With the improvement of maintenance skills, many agriculture aircraft agents have repaired machines on site. However, most of the in-warranty repairs involve responsibility ambiguity. Therefore, a method was introduced to analyze data on site to determine the responsibility and analyze the causes of aircraft accidents in front of users.

### 52. What does “Propulsion Output Reaches up to the Limit” Mean?

- 1. The aircraft is powered by the motor with forces in four directions during flight, including:
  - a) Roll torque (the force that causes the aircraft to turn left and right).
  - b) Pitch torque (the force that causes the aircraft to spin forward/back).
  - c) Yaw torque (the force that causes the aircraft to turn in the direction).
  - d) Lift (the force that counterbalances the gravity and stops the aircraft at a certain height).
- 2. In order to maintain the flight performance, safety, and stability of the aircraft when hovering or flying, the aircraft needs to be continuously adjusted through the motors to output the forces in these four directions.
- 3. When there is a large gap between the force required by the aircraft and the actual force provided by the motor, the propulsion output reaches up to the limit, which slightly reduces the flight performance, or threatens flight safety in a serious case.

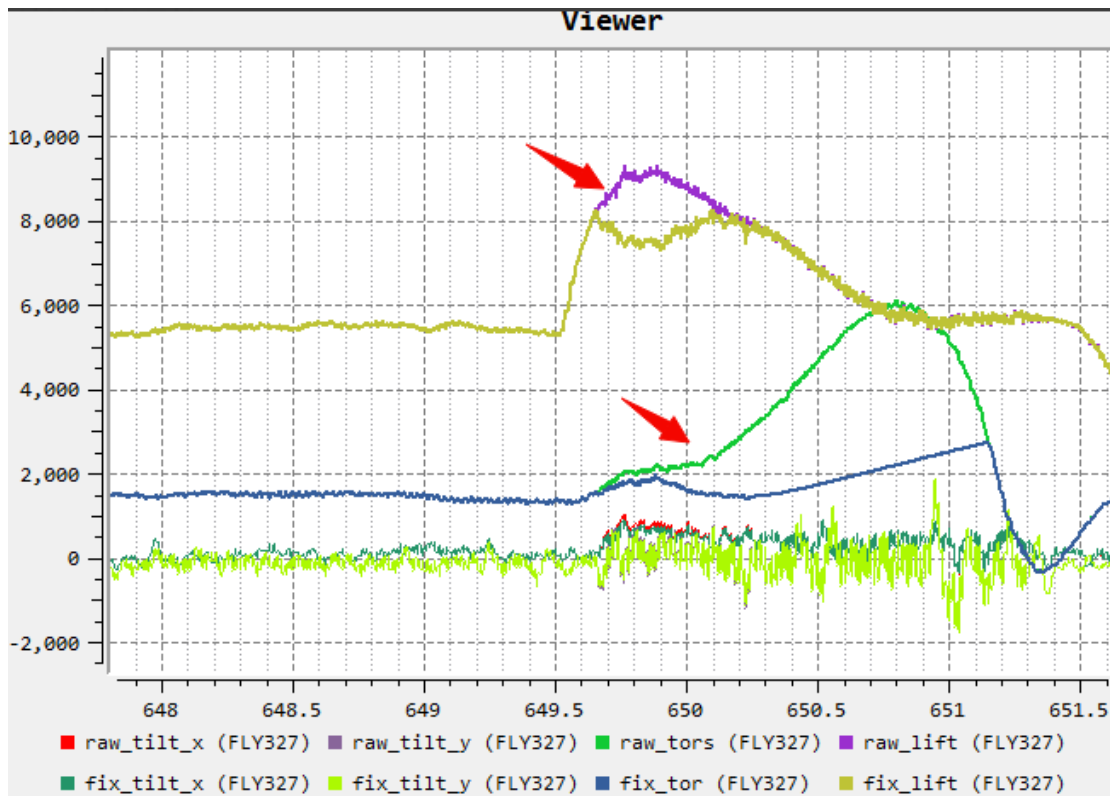
### 53. Example of the Situation that Propulsion Output Reaches up to the Limit

- 1. In the package `ctrl_allocation_debug`: `raw_tilt_x`, `raw_tilt_y`, `raw_tors`, and `raw_lift` correspond to the power forces required in four directions, which means that the aircraft needs these forces to be able to maintain normal performance.
- 2. These values are normalized, and their unit is 0.01%, that is, full scale -10,000 to 10,000





3. Fix\_tilt\_x, fix\_tilt\_y, fix\_tors, and fix\_lift correspond to the actual power forces output in four directions, which are the actual power forces the aircraft obtains, as shown below:



4. From the figure, the aircraft's actual force (fix) and the required force (raw) are equal before 649.5 seconds, and the power forces are in good condition.
5. After 649.5 seconds, fix is less than raw in the vertical direction and yaw direction, then the propulsion output reaches up to the limit.
6. By comparing packages
- raw\_tilt\_x - fix\_tilt\_x
  - raw\_tilt\_y - fix\_tilt\_y
  - raw\_tors - fix\_tors
  - raw\_lift - fix\_lift

It can be seen in which direction the aircraft propulsion output reaches up to the limit.

#### 54. Degree of the Situation that Propulsion Output Reaches up to the Limit

55.

- The aircraft propulsion output reaches up to the limit during the maneuvering operation. These conditions are not all bad, and most of the short-term phenomenon are safe.
- The real threat to flight performance and safety is that the propulsion output is limited for a long term, like for **more than 1 second**.
- If the propulsion output is limited for a long term, the phenomena of saturation in different directions can be expressed in the following table:

| Direction of saturation | Mild saturation | Severe saturation |
|-------------------------|-----------------|-------------------|
|-------------------------|-----------------|-------------------|

## Troubleshooting Guide for DJI Agricultural Products

|                  |                                                                       |                                 |
|------------------|-----------------------------------------------------------------------|---------------------------------|
| [1] Roll torque  | Severe overshoot in horizontal attitude                               | Aircraft flips over and crashes |
| [2] Pitch torque |                                                                       |                                 |
| [3] Yaw torque   | Bad hand feel if turning yaw, and severe overshoot if controlling yaw | Aircraft rotates out of control |
| [4] Lift         | Slight height loss, slow height gain                                  | Height loss of the aircraft     |

**Mild saturation:** Refers to the case where the fix is less than raw but greater than raw 80%; or the raw is less than 15% [greater than -1,500 and less than 1,500].

**Severe saturation:** Refers to the case where the fix is less than raw 80%, and the raw is greater than 15% [less than -1,500 or greater than 1,500].

4. As can be seen from the above table, the consequences are different for different saturation directions. Therefore, when propulsion output reaches up to the limit, the flight controller will deploy the power forces, and prioritize the power forces in the directions of [1] and [2]. This is a protection mechanism.
5. So from the performance of the whole aircraft, when power force saturation occurs, if:
  - a) **C1.** The aircraft's power is still sufficient, the flight performance of the aircraft is not affected.
  - b) **C2.** The aircraft's power is slightly less than the required power, the flight performance of the aircraft declines, the overshoot is intensified, and the hand feel becomes bad.
  - c) **C3.** The aircraft's power is obviously less than the required power, the aircraft turns yaw and spins out of control.
  - d) **C4.** The aircraft's power is significantly less than the required power, the aircraft turns yaw and spins out of control, and gradually falls to the ground.
  - e) **C5.** The aircraft's power is not enough to maintain the attitude, aircraft flips over and crashes.

[Remarks]: C1→C5 represent the degrees of power force saturation from mild to severe.

### 56. What to do at the Moment when the Propulsion Output Reaches Up to the Limit

1. For agriculture aircraft in five states of power force saturation, the recommended treatments are different.
  - a) **C1:** Operation can continue, and the app will not prompt any warning.
  - b) **C2:** It is probable for the app to prompt power force saturation. When the hand feel becomes bad, the effective control of the aircraft depends on the pilot's experience. It needs to land as soon as possible
  - c) **C3:** App prompts power force saturation. It is recommended to control the aircraft to land as soon as possible. Now, it is advisable to gently push the stick by no more than half of the stick range. No operation should be made on the yaw stick. In addition, it should be noted that although the aircraft is spinning out of control at a fast speed, command stick can still control the aircraft to fly in four directions in the air without any influence from the spinning. However, when this is happening the aircraft doesn't know which way it is facing, so it is necessary to push the stick carefully.
  - d) **C4:** App prompts power force saturation. It is recommended to control the aircraft to land as soon as possible. Now it is advisable to carefully push or even not to push the stick to allow the aircraft to spin on its own and land (damaging the landing gear) to

avoid greater losses due to flipping and crashing. **Never operate the yaw stick and push the throttle stick.**

- e) C5: Avoiding the Aircraft Falling
- 2. For the 1A/P agriculture aircraft, depending on the load, it will enter the C1 to C4 state when the power of one small arm is equivalently lost and will enter the C4 to C5 state when the powers of two small arms are equivalently lost.
- 3. In addition, it should be noted that when the aircraft is in the C3 (C4) state, if the user pushes the throttle by a large distance or forward/back, left/right, and the aircraft is (highly) probable to enter the C5 state.

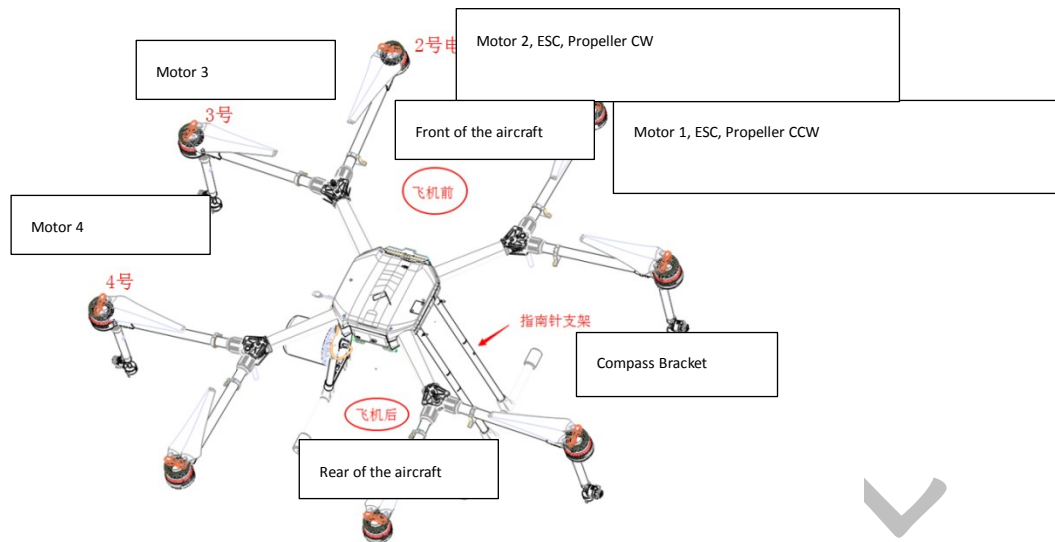
### 57. Cause of the issue Where the Propulsion Output Reaches Up to the Limit

- 1. Cause of power force saturation:
  - a) Propeller damage, loss.
  - b) Motor stalling.
  - c) ESC exposed to water damage or burnt out.
  - d) Arm structure break.
  - e) Battery voltage is abnormally low.
  - f) Excessive load.
  - g) The aircraft is shaking too much.
- 2. The above are common causes of power force saturation. When it occurs, it is possible to locate R1 and R4 first from the appearance, R5 through the battery voltage information, R2 (low speed, large current) and R3 (ESC disconnected or abnormal) from the ESC package, and R7 if the raw data in ctrl\_allocation\_debug is shaking too much.
- 3. Data analysis tutorial link: <https://bbs.dji.com/thread-150201-1-1.html>

### 58. How to Rescue the Aircraft if it Drops into Water?

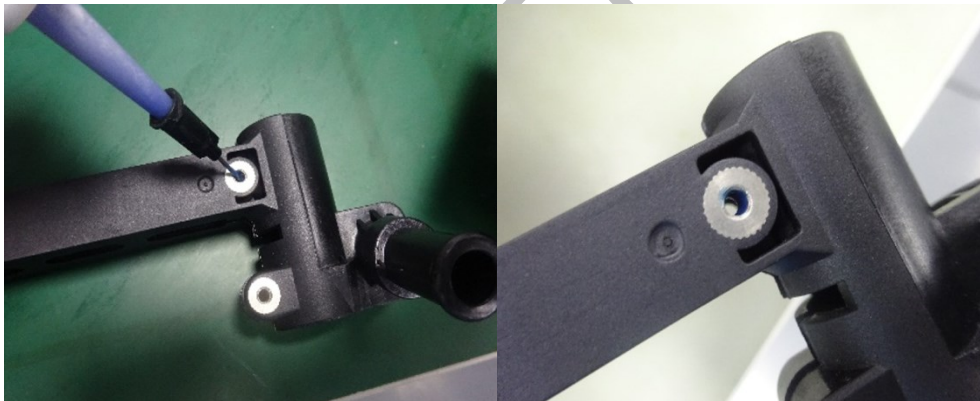
- 1. After confirming that the aircraft motor has stopped rotating, remove the battery from the aircraft to avoid more damage to the aircraft caused by the battery leaking.
- 2. Then check how the aircraft looks after falling into the water and try to keep other parts of the body away from the water. The ESC and the modules in the body have liquid contact indicators, which will turn red when they are exposed to water and cause the module board to be unstable.
- 3. Disassemble the upper shell and the parts stained with water, and check how they look. Any sand and dirt should be cleaned off quickly. Use a hair dryer or other equipment to air-dry the water on the aircraft, and then power on for detection. (If the conditions permit, it is best to remove the broken propellers and then power on for detection).

Image:

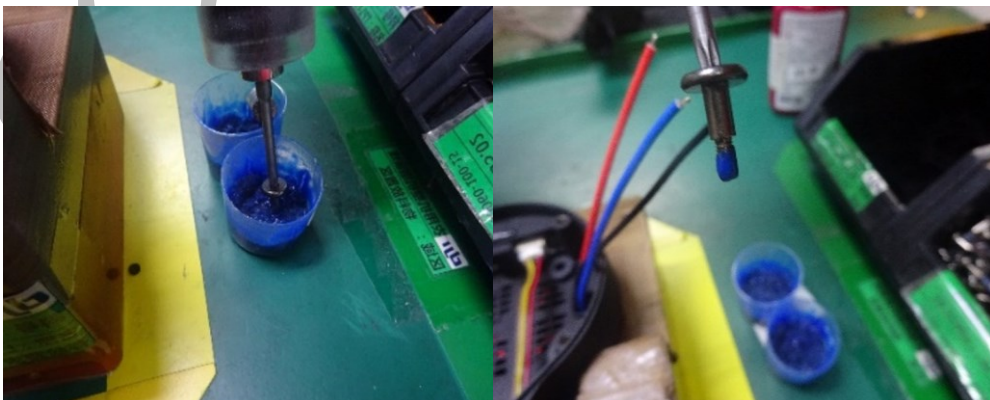


### 59. Guidance on Applying Glue to some Screws on the Fuselage

1. Mounting the landing gear beam:



2. Mounting the motors and motor bases:



3. Mounting the motor base cover ESCs:

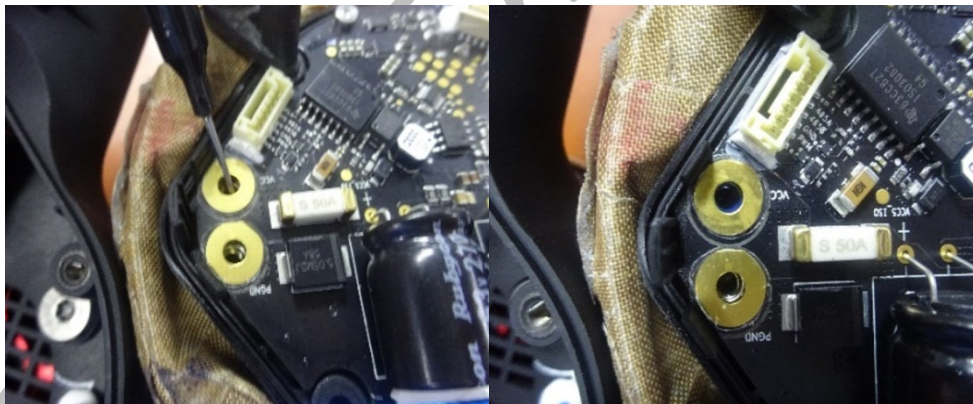




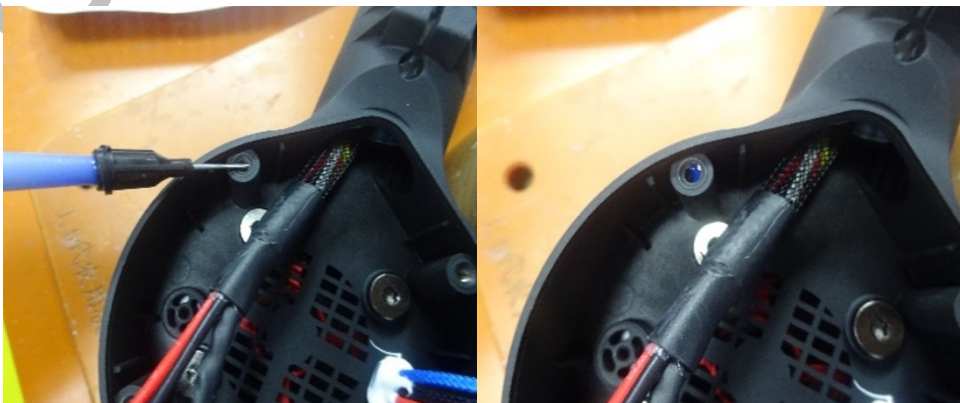
4. Mounting the elliptical tubes and T connectors:



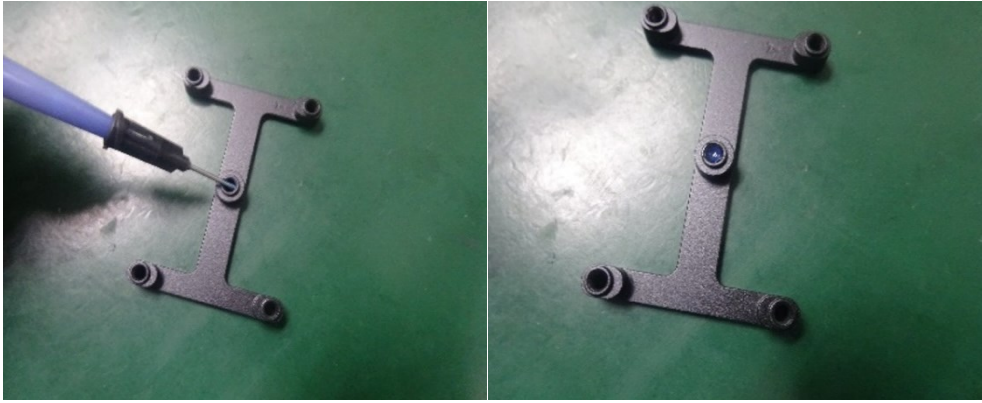
5. Mounting the motor components and elliptical tube assemblies:



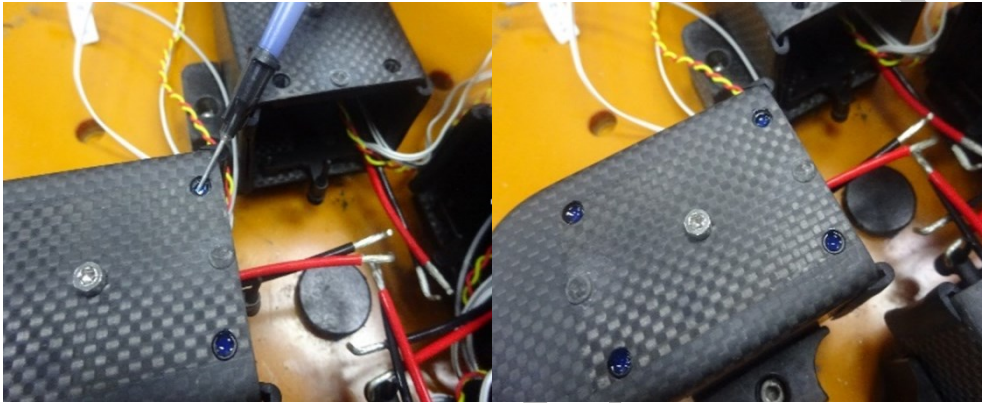
6. Tightening the ESC covers:



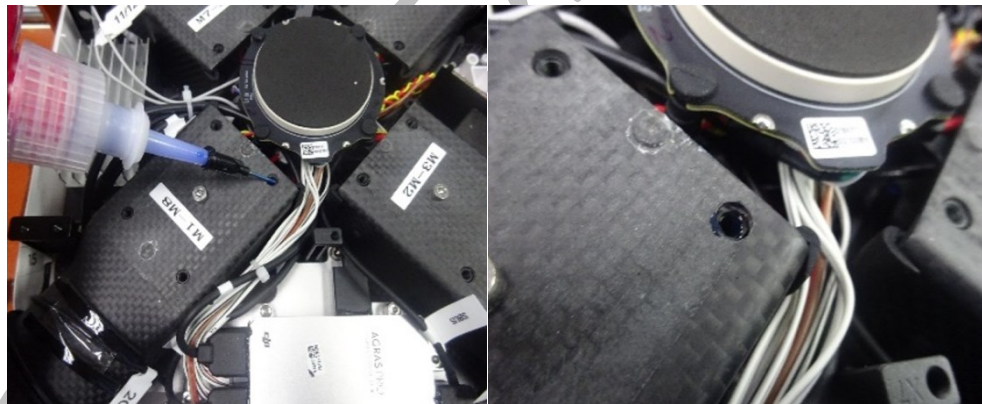
7. Mounting the single arm and elliptical tube aluminum pieces:



8. Mounting the lower carbon board:



9. Mounting the upper carbon board:

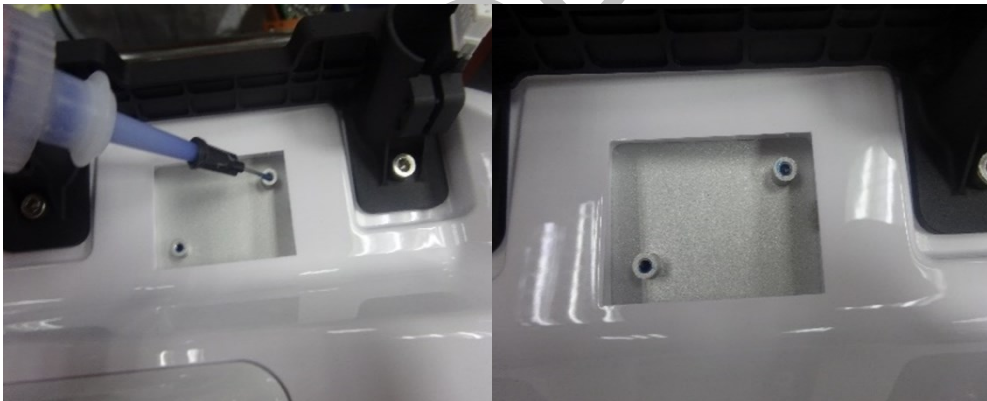


10. Mounting the motor bases and small arm carbon tubes:





11. Mounting the heat sink on the SDR lower shell:

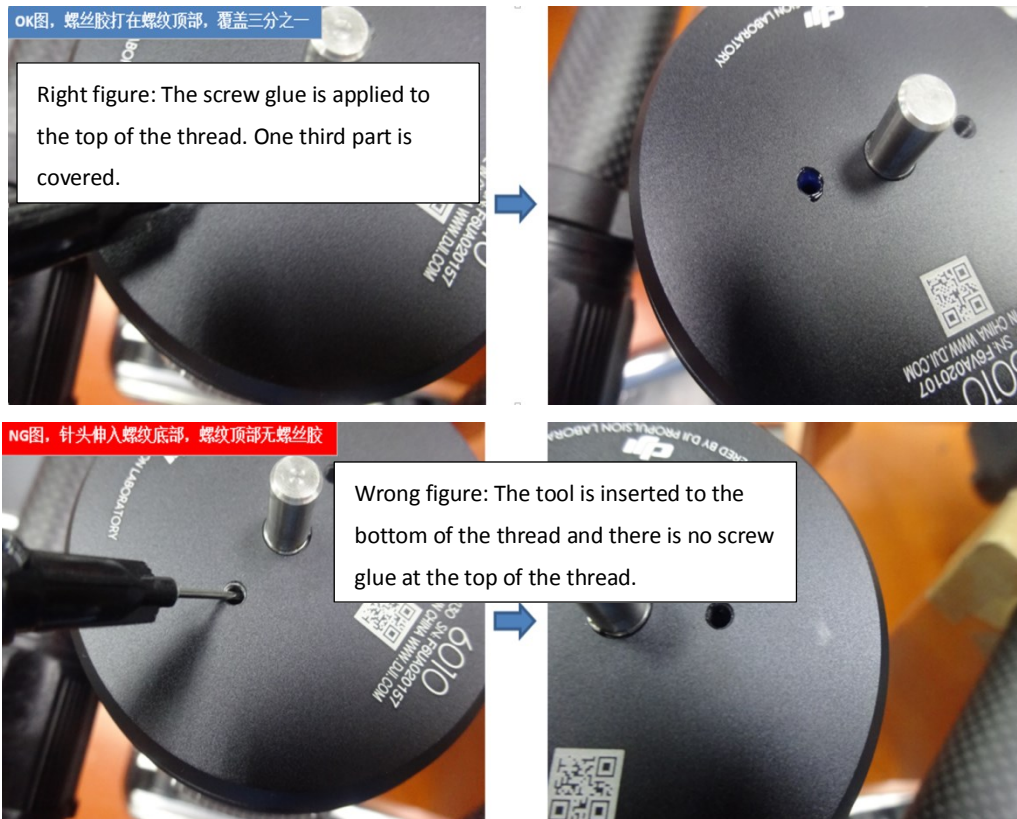


12. Mounting the propellers:



13. Example of applying glue to the screws:





[Remarks]: Put the tip of the tool into the threaded hole and apply the glue to the screw thread. It is necessary to ensure that the thread is glued from the inside to the outside. The thread should be fully immersed in the screw glue. The screw glue is applied to at least one-third of the threaded hole.

## VIII. MG Drone Repair and Maintenance

### 60. Remote Controller Maintenance

1. Check the stick mode, remote controller power level, stick neutral point, and flight parameter settings before starting.
2. Recalibrate the stick and confirm the stick orientation, if necessary (distinguish between American, Japanese, and Chinese preferred stick modes).
3. Avoid water and liquid from entering the remote controller during the operation (the remote controller lanyard is required for the operation).
4. Wipe the surface of the remote controller with a dry rag or alcohol after the process to keep it clean.
5. Avoid over discharging the remote controller battery.

## 61. Cleaning and Maintenance of the Whole Aircraft

### 62. Aircraft Body

The body is the main frame of the multi-rotor UAV. Note:

1. The aircraft is equipped with various electronic equipment, so water damage should be avoided.
2. Keep the outer shell closed. The agriculture aircraft often work in harsh environments, so it is necessary to check if the outer shell is closed before use.
3. Avoid placing things on the upper shell. The GPS device is often located in the middle of the aircraft, so things or thick stickers in this location may lead to a bad GPS signal.
4. Use a wet cloth to clean the aircraft, arms, motors, propellers, landing gear, spray tank surface and other parts. When cleaning, do not directly flush the motor, aircraft and other parts with hoses or high-pressure water guns. After cleaning, dry them to avoid corrosion by residual liquid inside the aircraft. (As the inside of the aircraft is not waterproof, it is advisable not to use a too wet towel for cleaning, to avoid moisture getting inside.)

#### 1) Landing Gear

Notes for landing gear:

1. Do not land the aircraft too fast. For weight reduction, multi-rotor aircraft are designed with lightweight landing gear. If the landing speed is too fast, it may cause excessive impact force and damage the aircraft.
2. Do not land on surfaces that may contain magnetic materials. As the compass of multi-rotor aircraft is usually installed inside the landing gear, if the landing gear lands on a surface containing magnetic material such as a surface containing a lot of steel bars, magnets, etc., the compass may be disturbed. In this case, the aircraft cannot fly normally.

#### 2) Spraying Operation Tank

In the idle season, the aircraft should be thoroughly cleaned to maintain the equipment performance and to improve the safety of the sprayed crop.

1. Add clean water to the tank, and spray the residual liquid in the tank and hose; open the lower outlet of the tank to discharge the remaining residual liquid, and remove the filter at the water outlet to clean.
2. Add water mixed with washing detergent to spray, and clean the spraying system.
3. Add clean water to clean the spraying system again.
4. Wet a cloth with clean water or water mixed with washing detergent, wring it out and wipe the aircraft.

### 3) Clogged Nozzle Cleaning

Improper operation and cleaning may cause irreversible damage to the nozzles. Note the following points when cleaning the nozzles:

1. Do not use hard materials such as iron wire and pins to poke the nozzle holes.
2. Acceptable tools: toothbrushes, toothpicks or other soft brushes. Remember that the cleaning tools should be much softer than the material of the nozzles.
3. High-pressure gas or clean water or ultrasonic cleaning may also be used to clean nozzles.
4. Cleaning of spray tank, water pumps, and pipeline: pour clean water into the spray tank after the operation is completed. Add an appropriate amount of detergent. Start the water pumps to clean the entire spraying system. Repeat 2~3 times. (a spray tank which contained herbicide should be cleaned more thoroughly to avoid phytotoxicity caused by secondary use).
5. Cleaning of nozzles and strainers: Clean the nozzles and strainers with fine-bristled toothbrushes. After cleaning, soak the nozzles and strainers in clean water. (Avoid the use of powders and high-concentration emulsion liquid.)

#### Notes:

1. Cleaning personnel should wear protective equipment such as masks and gloves before cleaning to avoid direct contact with chemicals.
2. Transportation: Try to lift only the big arm during transportation. Keep the transportation personnel from close contact with the agriculture aircraft to avoid chemical poisoning.
3. If herbicide is sprayed, wash it several times or replace the spraying system. Wear masks and rubber gloves when cleaning aircraft and other operation tools. The wet cloth must be wrung out before wiping the aircraft. Take note not to rinse with a large amount of water.

[Note]: The modules inside the aircraft are not waterproof. After long-term use, there will be a slight gap between the upper and lower shells of the aircraft. Summer is a rainy season. Try to keep the agriculture aircraft dry and avoid exposure to moisture which causes the instability of modules and boards.

### 63. Propulsion System Maintenance

1. Before the aircraft takes off, check whether the propellers are damaged or cracked. If any damage or crack is found, replace them. Pay attention to distinguish between CW and CCW when replacing the propellers.
2. Before the aircraft takes off, connect to the assistant to test motor rotation to check for noise or resistance. If any noise or resistance is found, replace it.
3. Check the motor inclination angle and propeller installation/service condition. If angle adjustment is involved, refer to the motor angle adjustment tutorial.

Link: <https://share.weiyun.com/5ydYyhJ> Password: ndvses

## 64. Battery Maintenance

1. When two batteries are charged at the same time, the output power of the charger is 2,400W (in express charge mode), and 1,200W in slow charge mode. Therefore, pay attention to the choice of power supply lines, so as not to burn out the circuit and cause accidents.
2. Each battery should be slow charged at least once a day to balance the voltage difference of cells (reason: If the batteries are always express charged, the battery cells may have a voltage difference. The increased voltage difference will decrease the battery life).
3. The batteries should not be over-discharged (voltage should not be lower than 2.75V). The aircraft should be landed before the low battery warning occurs, and the batteries should be cooled before charging.
4. Requirements for charging environment: temperature should not be too low, cool, dry and well-ventilated, and flat terrain; away from crowds, inflammable, and explosive materials. Try not to charge in high-temperature environments, to avoid the charger from frequently entering over-temperature protection (if the charger enters over-temperature protection, it must be cooled in temperatures ranging from 0 - 40°C for 30 minutes). If the batteries are not used for a long time, they should be fully charged using “storage mode” on the charger and placed in the explosion-proof box. The batteries should be placed in a dedicated warehouse, isolated from people, and checked at least once a month.
5. The batteries should be handled gently and checked for deformation and damage if they accidentally fall; regularly check the indicator light or the battery status displayed on the app to see if it is normal.
6. Incorrect use can cause the liquid to corrode the aircraft battery and its connector, resulting in short-circuits and the spontaneous combustion of the battery. The user must prevent pesticides from corroding the battery after charging, during flight and in storage, and cover the charging connector cap to prevent the power connector from being corroded by pesticides.
7. The MG drone pilots should regularly check the battery body, handles, cables, and power connector to see if the appearance is damaged, deformed, corroded, discolored, and broken and if the connection between the connector and the aircraft is too loose. At the end of each operation, use a dry cloth to wipe the battery surface and the power connector to ensure that there are no residual pesticides and the battery is not corroded.

## 65. Socket Connection Maintenance

### 4) Checking for Cable Sparks and a Solid Connection

Connecting the aircraft connector to the battery connector will produce sparks. After repeated sparking, the connector metal will be oxidized, and the metal part will be blackened. As a result, the heat generated at the end of the connector will increase, and the flight will be affected.

1. The connector must be fully inserted for connection before takeoff. (The battery needs to be inserted and unplugged quickly when it is installed on the aircraft end.)

2. If the metal part of the connector is found to be blackened, replace the connector promptly. (Battery end, aircraft end.)
3. If the metal part of the connector is found to have a smaller outer diameter, and if the fit is too loose, replace the connector promptly or enlarge the outer diameter. Keep the battery balance connector from touching water or liquid.

## 5) Transportation and Storage

1. Safety must be ensured during transportation. During the process, the aircraft should be fixed or placed in a special transport box to avoid bumps.
2. The batteries should be placed in an explosion-proof box or a special transport box and isolated from the people in the vehicle. Fire-fighting equipment, insulated gloves, and fireproof tongs should be kept in the operation vehicle. If the batteries smoke or are on fire due to careless operations, do not spray with water. Use a large amount of dry sand or a fire extinguisher to extinguish the fire.
3. The aircraft and batteries should be properly packed when stored for a long time. Requirements for storage environment: cool and dry place, away from flammable and explosive materials, away from fire. Fire-fighting equipment such as dry sand and fire extinguishers should be kept in the warehouse.

## 66. Inspection Operation

### 6) Water Pump

| Part       | Failure                                  | Cause Analysis                                          | Repair Solution                                                                    |
|------------|------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------|
| Water pump | No response when the pump is powered on. | 1. Check if the water pump cable is connected properly. | Ensure that the cable is properly connected.                                       |
|            |                                          | 2. Check if the water pump cable is damaged.            | Replace the water pump cable.                                                      |
|            |                                          | 3. Inspect by “spraying system test”.                   | Replace the corresponding defective parts.                                         |
|            |                                          | 4. Inspect for NG by “pump test”.                       | Replace the pump.                                                                  |
|            |                                          | 5. Check if the center board is damaged.                | Replace the center board.                                                          |
|            | No pressure value.                       | 1. Check if the water pipes are properly connected.     | Ensure that the water pipes are properly connected, and there is no water leakage. |

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|  |                   |                                             |                                  |
|--|-------------------|---------------------------------------------|----------------------------------|
|  |                   | 2. Inspect with the “spraying system test”. | Replace the corresponding parts. |
|  |                   | 3. Inspect for NG by “pump test”.           | Replace the pump.                |
|  | Leaking, rusting. |                                             | Replace the pump.                |

### 7) Radar

| Part  | Failure                                             | Cause Analysis                                                | Repair Solution                                    |
|-------|-----------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------|
| Radar | Not spinning.                                       | 1. Check if the radar cable is damaged.                       | Replace the radar cable.                           |
|       |                                                     | 2. Inspect with the “radar aging test”.                       | Replace the corresponding defective parts.         |
|       |                                                     | 3. Test turntable base modules alternately for spinning.      | If NG, replace turntable base modules.             |
|       |                                                     | 4. Check if the grating or grating disc is damaged.           | If NG, replace the grating or grating disc.        |
|       | Not spinning smoothly.                              | 1. Check if there is any object interference inside.          | Clear interference objects.                        |
|       |                                                     | 2. Inspect with the “radar smoothness test”.                  | If NG, replace turntable base modules.             |
|       |                                                     | 3. Inspect with the “radar grating disc test”.                | Replace the grating disc or grating.               |
|       | Altitude stabilization, obstacle avoidance failure. | 1. Check if the app has the radar switch turned on.           | Turn on the radar switch.                          |
|       |                                                     | 2. Inspect with the “radar aging test”.                       | Replace the corresponding defective parts.         |
|       |                                                     | 3. Inspect with the “radar module loopback sensitivity test”. | If NG, replace “RF board + digital board modules.  |
|       |                                                     | 4. Inspect with the “wireless power supply performance test”. | If NG, replace “turntable base” modules.           |
|       |                                                     | 5. Recreate the problem with the radar flight test.           | If NG, replace “RF board + digital board” modules. |
|       | Water damage                                        | Internal boards are oxidized, corroded, damaged.              | Replace the radar module.                          |

### 8) GPS and Compass Modules

| Part | Failure                       | Cause Analysis               | Repair Solution         |
|------|-------------------------------|------------------------------|-------------------------|
| GPS  | Signal lost.                  | Check if the GPS is damaged. | Replace the GPS module. |
|      | Failure to search satellites. | Check if the GPS is damaged. | Replace the GPS module. |



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|         |                                   |                                                                                                          |                                            |
|---------|-----------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------|
|         | Failure to search satellites.     | Check if the GPS cable is bad.                                                                           | Replace the GPS cable.                     |
|         | Unstable aircraft hovering.       | Check if the GPS is damaged while ensuring that the main controller and motor are in good condition.     | Replace the GPS module.                    |
|         | Water damage.                     | Liquid contact indicator has turned red.                                                                 | Replace the GPS module.                    |
| Compass | Compass calibration error.        | 1. Check if there is large interference in the surroundings.                                             | Calibrate in places with low interference. |
|         |                                   | 2. Large compass module value.                                                                           | Replace the compass.                       |
|         | No compass data.                  | 1. GPS damage.                                                                                           | Replace the GPS module.                    |
|         |                                   | 2. Compass damage.                                                                                       | Replace the compass.                       |
|         | Deviation from the straight line. | Check if the compass is damaged while ensuring that the main controller and motor are in good condition. | Replace the compass.                       |
|         | Water damage.                     | Compass board is exposed to water damage and oxidation.                                                  | Replace the compass.                       |

### 9) SDR Module

| Part       | Failure                                            | Cause Analysis                                                                            | Repair Solution                        |
|------------|----------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------|
| SDR module | Failure to link                                    | 1. Check if the remote controller and SDR have the latest firmware.                       | Update firmware to the latest version. |
|            |                                                    | 2. Check if the link cable is damaged.                                                    | Replace the link cable.                |
|            |                                                    | 3. Check if the link board is damaged.                                                    | Replace the link board.                |
|            |                                                    | 4. Check if the SDR antenna is damaged.                                                   | Replace the SDR antenna.               |
|            |                                                    | 5. Test SDR modules alternately.                                                          | Replace the SDR modules.               |
|            | Short image transmission or no image Transmission. | 1. Check if the remote controller and SDR have the latest firmware.                       | Update firmware to the latest version. |
|            |                                                    | 2. Check if the antennas and feeders of the SDR module and remote controller are damaged. | Replace antennas or feeders.           |

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|  |              |                                             |                          |
|--|--------------|---------------------------------------------|--------------------------|
|  |              | 3. Check if the FPV module is damaged.      | Replace the FPV module.  |
|  |              | 4. Inspect by “SDR attenuation test” If NG. | Replace the SDR modules. |
|  | Water damage | Liquid contact indicator has turned red.    | Replace the SDR modules. |

### 10) FPV Module

| Part       | Failure            | Cause Analysis                                                                             | Repair Solution                        |
|------------|--------------------|--------------------------------------------------------------------------------------------|----------------------------------------|
| FPV module | No image           | 1. Check if the remote controller and aircraft have the latest firmware.                   | Update firmware to the latest version. |
|            |                    | 2. Check if the FPV coaxial cable is damaged.                                              | Replace the FPV coaxial.               |
|            |                    | 3. Check if the FPV-center board connection cable is damaged.                              | Replace the FPV-center board cable.    |
|            |                    | 4. Inspect FPV modules alternately for damage.                                             | Replace the FPV module.                |
|            |                    | 5. Inspect SDR modules alternately for damage.                                             | Replace the SDR modules.               |
|            | Image Interference | 1. Inspect the remote controllers alternately to see if it is a remote controller problem. | Further, check the remote controller.  |
|            |                    | 2. Check if the SDR antenna is damaged.                                                    | Replace the SDR antenna.               |
|            |                    | 3. Inspect SDR modules alternately for damage.                                             | Replace the SDR modules.               |
|            |                    | 4. Inspect FPV modules alternately for damage.                                             | Replace the FPV module.                |
|            | Searchlight is off | 1. Searchlight is not turned on in the app.                                                | Turn on searchlight.                   |
|            |                    | 2. Check if the FPV-center board connection cable is damaged.                              | Replace the FPV-center board cable.    |
|            |                    | 3. Inspect FPV modules alternately for damage.                                             | Replace the FPV module.                |
|            | Water damage       | Liquid contact indicator has turned red.                                                   | Replace the FPV module.                |

## 11) Center Board Module

| Part         | Failure                                 | Cause Analysis                                        | Repair Solution                                           |
|--------------|-----------------------------------------|-------------------------------------------------------|-----------------------------------------------------------|
| Center board | Not able to read battery communication. | 1. Check if the power connector is defective.         | Replace the power connector.                              |
|              |                                         | 2. Inspect center board alternately for damage.       | Replace the center board.                                 |
|              | Abnormal power output.                  | 1. Check if the cable is bad.                         | Replace the cable.                                        |
|              |                                         | 2. Check if the socket is damaged.                    | Replace the center board.                                 |
|              |                                         | 3. Inspect voltage with the “center board link test”. | Replace the center board if the corresponding link is NG. |
|              |                                         | 4. Inspect the center board alternately for damage.   | Replace the center board.                                 |
|              | Not able to connect to DJI Assistant 2. | Check if the center board is damaged.                 | Replace the center board.                                 |
|              | Water pump is not working.              | Inspect with the “center board link test”.            | Replace the center board if the corresponding link is NG. |
|              | No data on the level gauge.             | Inspect with the “center board link test”.            | Replace the center board if the corresponding link is NG. |
|              | Spreader is not working.                | Inspect with the “center board link test”.            | Replace the center board if the corresponding link is NG. |
|              | Spinning radar is not working.          | Inspect with the “center board link test”.            | Replace the center board if the corresponding link is NG. |
|              | Water damage.                           | Liquid contact indicator has turned red.              | Replace the center board.                                 |

## 12) Motor/ESC

| Part | Failure | Cause Analysis                               | Repair Solution                  |
|------|---------|----------------------------------------------|----------------------------------|
|      |         | 1. Check if the cable is connected properly. | Connect the cable properly.      |
|      |         | 2. Check if it is linked.                    | Link with the remote controller. |

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|           |                                      |                                                                             |                                                                                                                |
|-----------|--------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Motor/ESC | All the motors cannot be started.    | 3. Check if each ESC firmware is normal.                                    | Update ESC firmware.                                                                                           |
|           |                                      | 4. Check if the app prompts an exception.                                   | Further, check the cause according to the prompt.                                                              |
|           |                                      | 5. Check if the LED status indicators of the No. 5 and 6 ESCs are abnormal. | Analyze the cause according to the status indicators (refer to the user manual for the indicator description). |
|           |                                      | 6. Check if the spare battery board is damaged.                             | Replace the spare battery board.                                                                               |
|           | A single motor cannot be started.    | 1. Check if the ESC firmware is normal.                                     | Re-burn the ESC firmware.                                                                                      |
|           |                                      | 2. Check if the motor is stalled.                                           | Replace the motor.                                                                                             |
|           |                                      | 3. Inspect by “ESC board sweep test”.                                       | If NG, replace the ESC.                                                                                        |
|           | Abnormal noise comes from the motor. | The motor shaft is deformed and worn.                                       | Replace the motor.                                                                                             |
|           | Motor stalling.                      | Motor deformation.                                                          | Replace the motor.                                                                                             |

### 13) Main controller

| Part            | Failure                                     | Cause Analysis.                                               | Repair Solution                                                       |
|-----------------|---------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------|
| Main controller | Not able to connect to the DJI Assistant 2. | 1. Firmware is missing.                                       | Re-burn the firmware.                                                 |
|                 |                                             | 2. Assistant version does not match.                          | Download the appropriate assistant version from the official website. |
|                 |                                             | 3. Computer driver installation failed.                       | Reinstall the assistant or change the computer.                       |
|                 |                                             | 4. Center board - main controller assistant cable is damaged. | Replace the assistant cable.                                          |
|                 | Update failure.                             | 1. Firmware is missing.                                       | Check the packet version and re-update.                               |
|                 |                                             | 2. Main controller - center board cable is damaged.           | Replace the cable.                                                    |

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|  |                                      |                                                                                                     |                                                    |
|--|--------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------|
|  |                                      | 3. Network is abnormal.                                                                             | Ensure that the network is smooth and then update. |
|  |                                      | 4. Avionics system is not fully connected.                                                          | Ensure that all modules are fully connected.       |
|  |                                      | 5. A normal main controller is changed to see if it is OK.                                          | Replace the main controller module.                |
|  | Main controller communication error. | Some function modules cannot be started or start with an error (modules are in good condition).     | Replace the main controller module.                |
|  | Pins are bent and broken.            | It is unable to output communication.                                                               | Replace the main controller module.                |
|  | Unstable aircraft hovering.          | 1. IMU calibration is abnormal.                                                                     | Recalibrate the IMU.                               |
|  |                                      | 2. When GPS module and motor are absolutely normal.                                                 | Replace the main controller module.                |
|  | Deviation from the straight line.    | 1. Compass calibration is abnormal.                                                                 | Recalibrate the compass.                           |
|  |                                      | 2. Check if the main controller is abnormal when the compass and calibration are absolutely normal. | Replace the main controller module.                |
|  | Water damage.                        | Liquid contact indicator has turned red.                                                            | Replace the main controller module.                |

### 14) Remote controller

| Part | Failure           | Cause Analysis                                          | Repair Solution                          |
|------|-------------------|---------------------------------------------------------|------------------------------------------|
|      | Unable to charge. | 1. Check if the remote controller's battery is damaged. | Replace the remote controller's battery. |
|      |                   | 2. Inspect alternately using a normal mainboard.        | Replace the mainboard.                   |
|      |                   | 1. Assistant version does not match.                    | Download the appropriate assistant       |

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|                   |                                             |                                                                          |                                                 |
|-------------------|---------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------|
| Remote controller | Not able to connect to the DJI Assistant 2. |                                                                          | version from the official website.              |
|                   |                                             | 2. Computer driver installation failed.                                  | Reinstall the assistant or change the computer. |
|                   |                                             | 3. Inspect back processing boards alternately.                           | Replace the back processing board.              |
|                   |                                             | 4. Inspect alternately using a normal mainboard.                         | Replace the mainboard.                          |
|                   | Failure to link.                            | 1. Check if the remote controller's firmware matches the air system.     | Update remote controller's firmware.            |
|                   |                                             | 2. Inspect alternately using a normal RF board.                          | Replace the RF board.                           |
|                   |                                             | 3. Inspect alternately using a normal mainboard.                         | Replace the mainboard.                          |
|                   | Connection lost within the specified range. | 1. Check if the antenna is damaged.                                      | Replace the antenna.                            |
|                   |                                             | 2. Inspect with the "remote controller coupling attenuation test" If NG. | Replace the RF board or mainboard.              |
|                   |                                             | 3. Inspect with the "remote controller SDR noise test"                   | Replace the RF board or mainboard.              |
|                   |                                             | 4. Inspect alternately using a normal RF board.                          | Replace the RF board.                           |
|                   |                                             | 5. Inspect alternately using a normal mainboard.                         | Replace the mainboard.                          |
|                   | Screen abnormal.                            | Check if the screen or coaxial cable is damaged.                         | Change the screen.                              |
|                   | Unable to connect to Wi-Fi.                 | 1. Check if the Wi-Fi antenna is damaged or not connected.               | Replace or connect the Wi-Fi antenna.           |
|                   |                                             | 2. Inspect alternately using a normal Wi-Fi board.                       | Replace the Wi-Fi board.                        |
|                   |                                             | 3. Inspect alternately using a normal mainboard.                         | Replace the mainboard.                          |



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|  |                                   |                                          |                              |
|--|-----------------------------------|------------------------------------------|------------------------------|
|  | All impellers and buttons failed. | Check if the converter board is damaged. | Replace the converter board. |
|--|-----------------------------------|------------------------------------------|------------------------------|

### 15) Battery

| Part    | Failure                                   | Cause Analysis                           | Repair Solution        |
|---------|-------------------------------------------|------------------------------------------|------------------------|
| Battery | Unable to read the battery's information. | Check if the smart connector is damaged. | Replace the connector. |
|         | Bulge, leakage.                           | Battery damage.                          | Replace the battery.   |
|         | Cell damage.                              | Battery damage.                          | Replace the battery.   |
|         | Unable to charge and start up.            | Battery damage.                          | Replace the battery.   |